GENERAL ELECTRIC COMPANY
DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GEK-36507

GENERAL INSTRUCTIONS

FOR

ADVANCED TURBINE SUPERVISORY INSTRUMENT SYSTEM

GENERAL ELECTRIC

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1.0 INTRODUCTION

These instructions describe the equipment supplied with an Advanced Turbine Supervisory Instrument (ATSI) system along with its installation, operation, and maintenance. The ATSI system is a microprocessor controlled instrument system for measuring, displaying and recording the many physical parameters and operating conditions of a steam turbine. The magnitude of these parameters is displayed in bar-graph form on a CRT and/or recorded on a magnetic tape.

Certain selected parameters are also displayed on a second CRT or selected page of a single CRT, in time-amplitude line-graph form (called trend lines) covering the past one-hour time period. This CRT or page is used also to display, upon request, up to two microcomputer generated time-amplitude graphs called slope lines each having a present value corresponding to its associated data line. Each slope is thereafter updated along the specified slope as though it were another data point.

Hereinafter, references to the bar and trend CRT displays will not differentiate between physical CRT's and a so-called page where the signals are switched and displayed on a single CRT.

The digital subsystems involved will be described in some detail. More specific instructions for each of the analog measurement channels are included in the publications for the various measurement channels, as listed in the TABLE OF CONTENTS of the system instructions. However, in place of recording data on a chart recorder as referenced in these publications, the 2V to 10V output from each channel is fed into the digital section to be displayed and/or recorded on magnetic tape. Also, listed in the TABLE OF CONTENTS are instructions covering the major components used in the system and applicable drawings.

2.0 DESCRIPTION

A typical ATSI system is shown in Fig. 1 and consists of the following parts

(1) the control cabinet housing the electronic equipment and the magnetic tape recorder.

(2) various detectors placed on the turbine to measure vibration, expansion, position, etc.

(3) one or two CRT's, as selected by the customer, to display the measured parameters in the form of bar-graphs and/or trend lines.

(4) a trend select unit which allows the operator to select up to 20 trend and 2 slope lines for display on the trend CRT.

(5) a vibration phase angle display panel.

(6) a teleprinter for communicating with the microcomputer and for making hard copies of the requested data.

All of this equipment operates from 115 VAC ± 10% at either 50 or 60 cycles ± .5 cycle as specified on the cabinet nameplate. If an approved uninterruptible power supply (UPS) is not available in the station, a separate UPS will be supplied with the system, as an option to provide power for all system components except the teleprinter. The UPS must be capable of supplying system requirements for 30 minutes after loss of the initial AC source. Instruction GEK-63393A provides specific requirements.

2.1 MEASURED VARIABLES

Table 1 summarizes the various parameters and operating conditions of the steam turbine that are monitored by the ATSI system, and indicates how they are displayed and/or recorded.

2.2 ACCURACY

The CRT displays are accurate to within ± 3% of full scale with a resolution of 0.5% of full scale. This includes any errors in the CRT and its graticule, the A/D converter, and associated circuitry. It does not include errors in the detectors and analog amplifiers. Their accuracy is specified in the publications for the various measurement channels.

2.3 CABINET

The cabinet is a ventilated NEMA 1 type enclosure with double front and back doors. The right bay is convection cooled while the left bay, where a dust-free atmosphere is required, is fan cooled and the air filtered creating a slight positive pressure to exclude dust.

The electronic equipment housed in the cabinet, Fig. 2 and Fig. 3, besides the logic power supplies and a magnetic tape recorder, details of which are covered under separately listed instruction books, is divided into two basic categories, analog and digital. Although the inputs to the digital system are analog and individual amplifiers are used for each thermocouple input, for purposes of explanation, the equipment in the right bay above the thermocouple (TC) amplifier box(es) will be referred to as the analog section, and the TC amplifiers and all equipment in the left bay will be referred to as the digital section.

As shown in Fig. 2, the typical cabinet includes (right bay from top to bottom) two DC power supplies, an AC power supply, a test panel, a number of modules which contain the

TABLE 1

MEASURED VARIABLES DISPLAYED/RECORDED

Type of Channel	Meter	Bar Graph Display	Bar Graph Alarm	Trend Display	Trend Slope	Record
Vibration	no	yes	yes	yes	no	yes
- turbine	no	yes	yes	no	--	yes
- generator & exciter						
Vibration Phase Angle*	yes	no	--	no	--	no
Eccentricity						
- turbine stantard detector	no	yes	no	no	--	yes
Expansion						
- shell	no	no	--	no	--	yes
- differential	no	yes	yes	yes	no	yes
- rotor	no	no	--	no	--	yes
Speed	no	no	--	no	--	yes
Peak Speed	no	no	--	no	--	yes
Position						
- Control Valve	no	no	--	no	--	yes
- Nuclear Bypass Valve	no	no	--	no	--	yes
Steam Temperature						
- throttle	no	no	--	no	--	yes
- 1st reheat	no	no	--	no	--	yes
- 2nd reheat	no	no	--	no	--	yes
Water Detection Temperature	no	yes	yes	no	--	yes
Steam Seal Temperature						
- header	no	yes	no	no	--	yes
- 1 p. packing	no	no	--	no	--	yes
Exhaust Hood Temperature	no	no	--	no	--	yes
Exhaust Hood Pressure	no	no	--	no	--	yes
Shell Metal Temperature						
- S.V. inner surf.	no	yes	yes	no	--	yes
- S.V. outer surf.	no	yes	yes	no	--	yes
- C.V. inner surf.	no	yes	yes	no	--	yes
- C.V. outer surf.	no	yes	yes	no	--	yes
- 1st stg. inner surf.	no	yes	yes	yes	yes	yes
- 1st stg. outer surf.	no	yes	yes	no	--	yes
- 1st rh. bowl inner surf.	no	yes	yes	yes	yes	yes
- 1st rh. bowl outer surf.	no	yes	yes	no	--	yes
- 2nd rh. bowl inner surf.	no	yes	yes	yes	--	yes
- 2nd rh. bowl outer surf.	no	yes	yes	no	--	yes
Bearing Metal Temperature						
- journal brg.	no	no	--	no	--	yes
- thrust brg.	no	no	--	no	--	yes
Acceleration	optl.	no	--	no	--	no
System Status						
- d.c. oil pump running	no	no	--	no	--	yes
- lube press. low	no	no	--	no	--	yes
- circuit breaker open	no	no	--	no	--	yes
- turning gear engaged	no	no	--	no	--	yes
- rotor stopped	no	no	--	no	--	yes
Generator load (MW)	no	no	--	no	--	yes

*Selectable-channel type

analog printed wiring boards, and sitting on the floor, a box housing up to six therocouple amplifier boards each containing eight individual amplifiers. The left bay (top to bottom) contains a circuit breaker panel, magnetic tape recorder, two logic modules and a logic power supply panel.

The logic modules are designed for printed wiring boards (PWB's) with 86 pin connectors which plug onto pins on another larger PWB called the backplane board which covers the back of the module. This backplane board distributes the signals and power to the boards in the module. All outgoing and incoming signal connections are made to the back of the logic modules via connectors which plug onto the same pins as the PWB's, but which are protruding through the backplane board. Refer to the internal connection diagram for the exact number of boards used in a particular system and to the system elementary for signal routing between the logic boards.

A typical analog module is shown in Fig. 4. The circuit boards plug into receptacles toward the back, inside the module. The number of modules used varies with different systems.

2.4 ANALOG OUTPUTS

The following outputs are provided from each of the analog measurement channels (except for acceleration and vibration phase angle) and are wired to outgoing terminals for use as required:

1. One to 5 ma full scale, for indicating instrument. Permissible load resistance 0 to 1000 ohms, except for certain measurement channels where 2000 ohms maximum is permitted (see instructions for the particular measurement channel involved).

2. Eight to 40 millivolts, for digital computer and/or data logger. Source impedance 40 ohms. Connected load (computer input impedance) to be 5000 ohms or greater.

3. Two-tenths to 1.0 ma full scale, for chart recorder. Source impedance 9950 ohms. Recommended recorder input impedance, 50 ohms.

4. Two to 10 volts, with output limited by a 2000 ohm $\pm$ 0.1 percent series resistor. This output is for special purposes, and is only wired to outgoing terminals when specified. Maximum permissible current drain 2 ma at full scale (10 volts) output voltage. (Position channels, having two of these special 2 to 10 volt outputs, are limited to 1 ma from each, if both are being used.)

Neither side of the 1 to 5 ma output is at ground potential and, therefore, neither side of the connected load (milli-ammeter) may be grounded.

The negative side of the 8 to 40 mv output is grounded within the cabinet, and neither side may be grounded at the computer.

The negative side of the 0.2 to 1.0 ma output is grounded within the cabinet, and neither side may be grounded at the recorder.

NOTE: All the measurement channels have all the above outputs except for the acceleration channel and vibration phase angle. The acceleration channel has only one output, -5 to 0 to +5 ma, center scale. The vibration phase angle outputs are fully described in the applicable instructions.

In addition to the above outputs, each channel has a 2 to 10 volt full-scale output used internally as an input to the digital section and if required, an alarm board.

2.5 ALARM AND TRIP

Both the analog and digital sections have provisions for actuating alarms via relay contacts. Only the analog circuits will be used for tripping the turbine. In addition to the relay contacts, the digital section has provisions for flashing the bar-graph display channels when an alarm point is reached.

There is no simultaneity of analog and digital alarms, rather the analog alarms are set higher than the digital to act as backup. The digital alarm limits are set via eight-pole switches located on the PROM board(s). Table 2 summarizes the conditions that may be alarmed by the digital section. The system manufacturing instructions (MI) gives the settings for a particular unit.

In the analog section most systems include alarm boards for vibration channels, to actuate an annunciator, start indicator lights or other alarm device in case the vibration should exceed the preset limits. A trip board is usually included, connected in such a way that if any one or more vibration channels exceed their alarm set-points, the trip contacts will be actuated. These trip contacts may nor may not be connected to trip the turbine in case of excessive vibration, depending on the recommendation of the turbine manufacturer and the preference of the user. Systems may also be connected so the trip contacts will be actuated by excessive differential expansion, as well as vibration. The trip circuit will automatically be disabled is any of the vibration amplifier boards, vibration alarm boards associated with the trip

TABLE 2

DIGITAL SECTION ALARM SUMMARY

Type of Channel	Condition	Setpoint Adjust	Bar Graph Flash	Alarm Contact	
Vibration	High	Common to all Channels	Individual channel	Common to all channels	*
Differential (Exp.)	Long rotor	Separate for each channel	Individual channel for either high or low	Common to all channels	*
	Short rotor	Separate for each channel			*
Water Detection Temperatures	High difference between pairs	Common to all pairs	Both temp of a pair	Common to all pairs	*
Shell Metal Temperatures	High difference between pairs	Common to all pairs	Both temp of a pair	Common to all pairs	*
Journal Bearing	High	Common to all channels	None	Common to all	**
Thrust Bearing Metal Temperature	High	Common to all channels	None	Common to all channels	**
Digital Section	Loss of power supply	NA	NA	Common to all listed malfunctions	***
	Not recording on tape				
	Not sending to CRT				
	Not cycling through program				
	End of Tape				
	Teleprinter BAD status (not available if current interface is used)				

* - Contact #1
** - Contact #2
*** - Contact #3

function or the trip board is removed, or if there is a loss of either of the DC power supplies or if the AC power should fail. The tripping capability will be automatically restored about 45 seconds after the abnormal condition is restored to normal.

A TRIP switch is included for manually disabling the trip circuitry. Also, a trip supervisory relay is included to provide a warning whenever the trip circuit is disabled.

Alarm boards are usually included also for the differential expansion channels and, optionally, may be included for eccentricity, speed and other channels.

2.6 CONTACT RATINGS

1. Contacts of Alarm Boards

   One set of Form C contacts (normally open-normally closed combination) for each alarm relay, rated as follows:

   1/2 amp maximum at 120V AC, or
   1/2 amp maximum at 125V DC, or
   1/4 amp maximum at 250V DC.

2. Trip Supervisory Contacts

   One set of form C contacts (normally open-normally closed combination) rated the same as the alarm board contacts.

3. Low Output Alarm Contacts

   One set of form C contacts (normally open-normally closed combination), rated the same as the alarm board contacts.

4. Trip Contact

   One normally open contact rated 1 amp maximum, at 125V DC or 120V AC.

5. Digital Alarm Contacts

   Form C contacts rated a minimum of 1 amp at 125V DC or 120V AC.

2.7 CALENDAR CLOCK

The calendar clock board (clock board) located in logic module #1 along with its readout board, provides a visual display at the cabinet of the year, day of year, hour, minute, and second. It also provides this information to the system in digital form via buffers on the A/D board, to be processed and displayed on the CRT, as well as recorded with each log of data on magnetic tape.

The clock board does not provide timing or sync pulses to the microcomputer system. Such clock pulses are generated within each of the three main digital subsystems and synchronized as required.

Leap year is indicated on the readout via the decimal point in front of each numeral representing the year, and the clock will automatically count 366 days before being reset to day 1.

The clock readout board is equipped with eight switches for controlling and setting the clock. They are as follows:

1SW - Set Fast
2SW - Set Slow
3SW - Run/Hold

Hour, Min, Sec

4SW - Reset Day-Year
5SW - Set Day Fast
6SW - Set Day Slow
7SW - Set Year Fast
8SW - Set Year Slow

All of the above switches are momentary except for the RUN/HOLD switch which is maintained.

2.8 RECORDER AND MAGNETIC TAPE

The recorder is a synchronous digital magnetic tape unit (tape deck) that, with proper external formatting control, is capable of reading and writing IBM compatible tapes. The unit reads after writing and has the following additional specifications:

Input Voltage:	115V ± 10%, 50/60 Hz	
Tape Speed:	12.5 In./Sec	
Fast Speed:	160 In./Sec	
Number of Tracks:	9	
Recording Mode:	NRZI	
Recording Density:	800 bits/in.	
Reel Diameter:	7 in. to 10 1/2 in.	(limited to 600 ft. reels for this application)

For detailed instructions on the recorder, refer to its separate instruction book as listed in the TABLE OF CONTENTS of the system instructions.

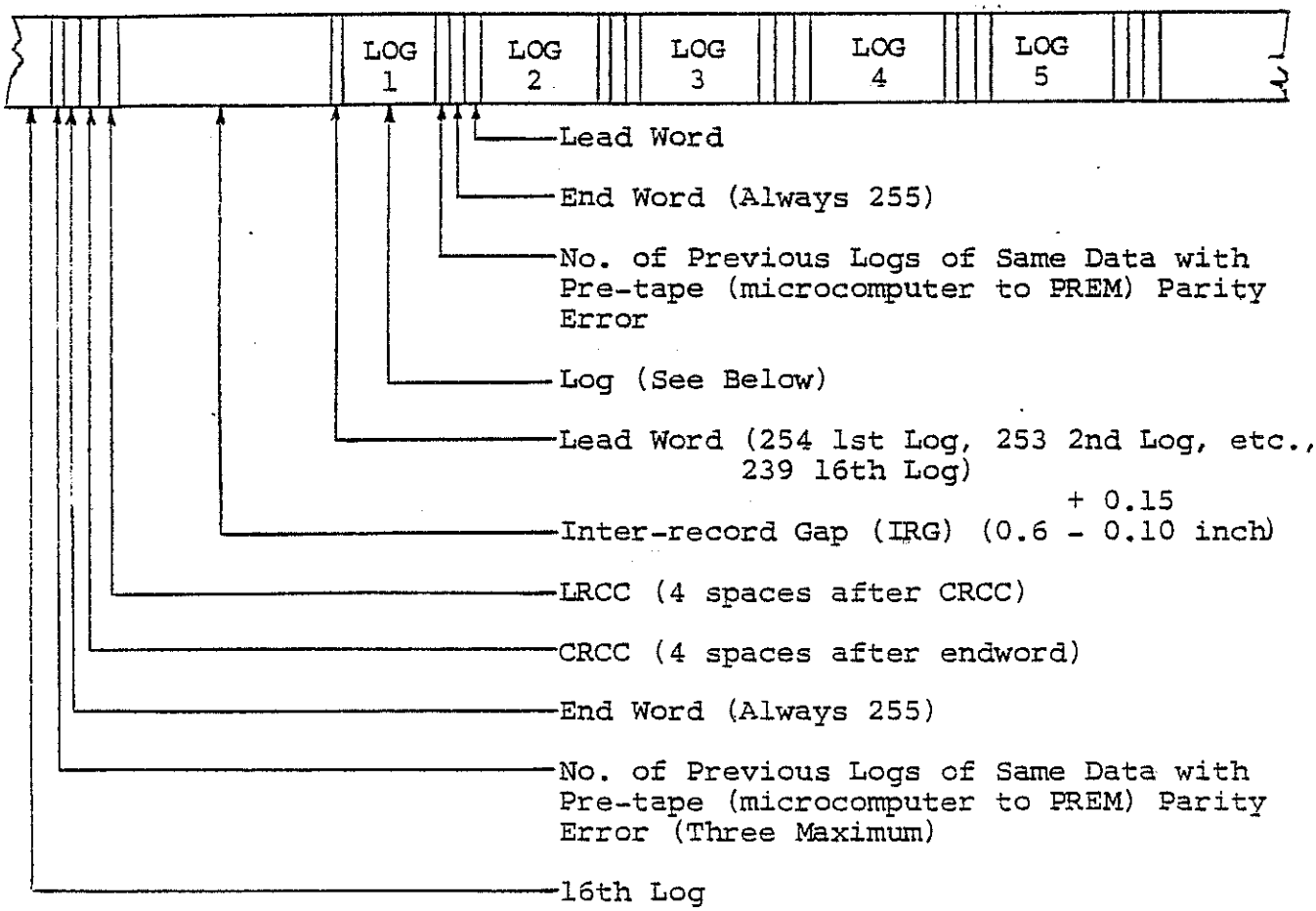
It is recommended that computer grade tape, with oxide coating on the back, 3M Company's Scotch Brand 700 Blackwatch type be used (600 ft. reel only). Each reel may hold 3-5 months of data depending on the number of alarms encountered and the system size. One reel is supplied with the equipment. Additional reels may be ordered from the tape manufacturer

or through the nearest Installation and Service Engineering office of General Electric Company.

The use of any other type tape, without the approval of General Electric Company, will absolve General Electric Company of all responsibility for any misoperation or data retention in connection with incompatibilities betweeen the system and the magnetic tape.

2.9 RECORDING FORMAT

Data is recorded on tape, in blocks of 16 logs each, using the non-return to zero invert format. The following is the buildup of each block:



Each log is built up as follows:

Year
100's of Days + Leap Year
Days
Hours
Minutes

Seconds
Turbine No. (MSB)
Turbine No. (LSB)
Data (Refer to the external connection diagram, listed in the TABLE OF CONTENTS of the system instructions, for the order in which the parameter data is recorded and their range. All readings are clamped at full scale and will not register higher.

The time interval between each complete log of data is one of two values, depending on conditions described below, except for immediate logs. The longer interval is 5 minutes; the shorter is 30 seconds. The system automatically goes to the 30-seconds interval for 24 minutes after initial startup or restart after power off and during the presence of any one of the alarm conditions listed below. If any one of the "System Status" inputs listed in Table 1 changes state, a log will be taken immediately.

Logs are taken at 30-second intervals upon an alarm condition as shown below.

Type of Channel Alarmed	Time Interval at Fast Log Rate
Vibration	24 minutes
Differential (Expansion)	Take log immediately, then slow rate
Water Detection Temp	Slow rate only
Shell Metal Temp	Slow rate only
Journal Bearings Metal Temp	24 minutes
Thrust Bearing Metal Temp	24 minutes

When the following conditions are present, there will be no fast logging:

1. Turning gear engaged
2. Tape recorder unavailable to receive data
3. Rotor stopped
4. Low tape alarm is on
5. Generator is ON LINE.

2.10 MAGNETIC TAPE DATA PROCESSING

The ATSI tapes are IBM compatible and conform to American National Standard X3.22-1973, Recorded Magnetic Tape for

Information Interchange (800 CPI, NRZI). To process these tapes other than by use of the ATSI system, the following equipment would be required and may be purchased from several vendors.

1. 9 track tape reader
2. Controller
3. Computer with memory
4. Line printer or CRT

In addition to the above equipment, the development of a computer program would be required for reading the data into memory.

2.11 TAPE INFORMATION

TAPE	9 track (8 data tracks, 1 parity track)
DENSITY	800 bpi
FILE NUMBER	1
LABEL TYPE	no labels
RECORD FORMAT	blocked fixed-length records
LOGICAL RECORD LENGTH	varies with turbine, up to 111 bytes
BLOCK SIZE	varies with turbine, up to 1776 bytes
PARITY	odd
DATA LISTING	See recording format and refer to listing on the external connection diagram supplied for the system.

The initial gap between the trailing edge of the BOT marker and the first block of data is 5 inches.

The time, date, and turbine number data is recorded as two BCD digits per byte except for 100's of days which, additionally has bit 3 set during leap year. All other data points except statuses are calibrated for a binary count of 200 at full scale. Status points equals 000 if count is < 10 otherwise they equal 001.

9 Track Identification

Track	Bit	Weight
1	3	2^2
2	1	2^0
3	5	2^4
4	P	Parity
5	6	2^5
6	7	2^6
7	8	2^7
8	2	2^1
9	4	2^3

For additional information, refer to the American National Standard referenced above.

2.12 DATA MONITOR(S)

The data monitor(s) in the system is designed for panel or rack mounting with standard 19 in. rack spacing. Refer to the outline drawing supplied for exact dimensions.

The monitor(s) includes a CRT tube with medium-presistence, yellow-green phosphor and conventional controls and circuitry necessary to accept a standard video signal over a coaxial line.

CAUTION

THE BRIGHTNESS OF THE DISPLAY MUST BE KEPT AT A MEDIUM TO LOW SETTING TO PREVENT PATTERN BURNS ON THE FACE OF THE CRT.

The monitor(s) will exhibit appropriate electronically generated graticules and will have appropriate legend and scale markings, either by means of a separate overlay placed on the face of the CRT or by electronic generation.

Scale markings for the trend display are as follows: on the right side a scale of 0 to 100% in the vertical direction is used, for reading parameters other than temperatures. On the left side an appropriate temperature scale is used. One hundred percent represents the same full scale value for that parameter as is shown on the bar CRT. The horizontal scale is marked 0 to -60 minutes with "0" indicating present time, at the right.

2.13 TREND SELECT UNIT

The trend select unit (shown in Fig. 5) is designed for use in conjunction with the trend display and should be mounted in easy reach of the operator. The unit has two basic functions. First, the unit contains the pushbuttons and

necessary circuitry to allow the operator to select one or more of the predetermined parameters in any combination to appear on the trend CRT in time-amplitude line graph form. Pushing a button merely permits or inhibits the display of that line on the screen and does not affect the memory that stores that line's history.

Second, the unit has pushbuttons and circuitry that are used to request up to two microcomputer generated time-amplitude graphs called slope lines. These lines are normally used in conjunction with the first stage and first reheat bowl inner shell metal temperatures and allows the operator to compare the actual parameter value against a selected (preferred) rate of change in value.

The trend select unit is designed for panel or rack mounting with standard 19-inch rack mounting dimensions. Refer to the outline drawing supplied, for exact dimensions.

In systems where a single CRT is specified, this unit contains a momentary pushbutton which allows the operator to switch the CRT display from the bar graph data to the trend data. The trend display will only be maintained for as long as the button is depressed. A mechanical latch is provided to hold the button depressed but normally should not be used.

3.0 RECEIVING, HANDLING AND STORAGE

3.1 GENERAL

Upon receipt of the ATSI shipment, examine it immediately for any damage that might have occurred in transit. If damage is found, a damage claim should be filed immediately with the transportation company, and the nearest Installation & Service Engineering office of General Electric Company should be notified.

If the equipment is to be stored prior to installation, it should be left packed in the original packing material and containers. Store in a dry area which is free of corrosive vapors and where the equipment will not be subjected to excessive shock, vibration, or temperature extremes.

The recorder which mounts in the cabinet is shipped separately. It is recommended that it not be mounted until after the cabinet has been installed and completely wired. Special precaustions should be taken to insure that the recorder be kept clean and dust free at all times.

3.2 CARE OF PRINTED WIRING BOARDS

Printed wiring boards (PWB's) are used throughout the system. Avoid rough handling of the boards. Handle the boards by the edges. Avoid touching the printed wiring portions of the

boards with the fingers, as this may cause leakage problems under humid conditions or, in the case of PWB's with MOS devices, a static discharge that may damage the device.

CAUTION

1. If boards are removed from the logic modules for any reason, they should be placed on a grounded conductive surface.

2. When placing boards on or removing them from the conductive surface, the handler should first ground himself before touching the board so that there is no possibility of a static discharge through any component on the PWB.

3. When storing or shipping any digital PWB, a conductive envelope or wrapper is desirable. Nonconductive plastic wrapping must be avoided.

Dry weather (relative humidity less than 30%) tends to multiply the accumulation of static charges on any surface. Therefore, in a low-humidity environment, the preceding handling precautions take on added importance.

If a board must be removed from its module, be certain to return it to exactly the same position from which it was removed. For the correct position of the boards, refer to the internal wiring diagram included in the system instructions.

CAUTION

TO AVOID DAMAGE TO THE PRINTED WIRING BOARDS, IT IS NECESSARY THAT POWER BE TURNED OFF THE DIGITAL SECTION BEFORE BOARDS ARE REMOVED FROM OR INSERTED INTO THE LOGIC MODULES.

3.3 CARE OF MAGNETIC TAPE

Data recorded on magnetic tape will not deteriorate and will remain unchanged unless actually altered by an external magnetic field, but physical damage will greatly affect performance. Excellent performance with a minimum of hard (uncorrectable) errors should be expected when storing tapes under the following conditions. Tapes should be stored only at slowly changing temperatures of 0 to 120°F (-17.8 to 48.9°C), relative humidity of 20 to 80%, conditioned (24-hour wait period in normal environment) before usage, and not exposed to hard mechanical shocks. Back coated tapes are used to reduce the effects of static charges on the tape (attraction of dust particles) but precautions must be taken to maintain a clean, dust-free environment during usage and storage.

Cleaning of the tape unit each time a reel is changed is a must to prevent spreading contamination from one reel of tape to another. Tape contamination caused by fingerprints can be reduced by remembering not to touch the tape unnecessarily. Refer to the maintenance section of the recorder instruction book for proper cleaning procedures.

When a reel is removed from the unit to be stored, secure the end of the tape by use of adhesive tape or a small piece of sponge jammed into reel. Replace the plastic edge seal ring received with the reel, place reel in a clean plastic bag and seal by folding over and applying adhesive tape. In lieu of the above, a reel container may be used.

To prevent any increase in error growth during storage, it is recommended that recorded tapes be exercised (unwound, cleaned, and rewound) once a year.

4.0 INSTALLATION

The electronic equipment is for indoor, non-tropical installations, and must be located where they will not be subjected to water, moisture or corrosive vapors and where they will be reasonably free from dust and dirt.

4.1 CABINET

The ATSI cabinet may be located in any reasonably dust and dirt free air conditioned area of the plant, and should be situated so both the front and the rear doors can be opened fully for servicing. The doors to the cabinet and the recorder inside, must be kept closed at all time except during brief periods of routine maintenance.

CAUTION

THE RECORDER MUST NOT BE ALLOWED TO OPERATE IN AN ATMOSPHERE OF DUST OR LINT AS DAMAGE TO THE RECORDING HEAD WILL RESULT.

The electronics in the cabinet are capable of operating over a temperature range of 35° to 122°F (2° to 50°C) and at a relative humidity of 15 to 95% without condensation. Since most commercially available magnetic tapes are limited to 95°F for operation and 120°F for storage, as is the tape recommended for use in ATSI, the ambient temperature range for the cabinet must be derated to 40 to 90°F (4.4 to 32.2°C), with a relative humidity of 20 to 80%. Within this recommended ambient range, the temperature should be maintained within ± 5°F and the relative humidity should be kept constant to within ± 10%.

The cabinet must be located so the length of the wiring from the cabinet to the CRT data monitors and teleprinter will not

exceed 200 ft. It is recommended that the length of wiring from the detectors, on the turbine, to the cabinet not exceed 650 ft., but in any case the loop resistance must not exceed 35 ohms.

The cabinet is provided with an opening in the bottom to permit wiring to be brought in from ducts or conduit under the floor. However, when preferable, wiring may be brought into the top of the cabinet instead. In the latter case, it will be necessary to remove a plate from the top of the cabinet and drill or cut holes for cable or conduit entrance, then replace the plate. To prevent dust and dirt from entering through the bottom of the cabinet, it must be set flat on the floor or otherwise sealed.

The magnetic tape recorder which mounts to a sliding frame in the left bay of the cabinet has been shipped separately. It should not be installed until after all connections to the cabinet have been made as the recorder must be kept as dust free as possible. It is recommended that the original packing carton be retained for future use in the event the recorder must be returned for servicing.

To mount the recorder, remove the two bolts retaining the sliding frame in position and move it forward until it hits its two stops. Mount the recorder to the frame with the hardware provided and return the slide to its original position, replacing the two retaining bolts. (Refer to the cabinet's internal wiring diagram for making the proper connections to the recorder which is by means of connectors.) The recorder's power cord plugs into the 115 VAC receptable "CO" mounted on the side wall of the cabinet near the back of the recorder.

4.2 DATA MONITOR(S) AND TREND SELECT UNIT

The data monitor(s) and trend select unit is designed for panel or rack mounting with standard 19 in. rack spacing. These units should be located in a place convenient for the operator, with the trend select unit in easy reach and preferably directly under the trend monitor. The distance from the monitor(s) and trend select unit to the cabinet must not exceed 200 ft.

To mount the monitor(s), first remove the CRT chassis from it's rack hanger by removing the four retaining screws through the bottom of the rack. Mount the hanger. Then slide the chassis back into the hanger and secure by the two screws toward the back. (Refer to the outline drawing supplied for the unit.)

To mount the trend select unit, remove the front cover by loosening the two cover-retaining socket-head screws on each side of the unit. Then mount the unit and replace the cover (Refer to the outline drawing supplied for the unit.)

4.3 TELEPRINTER

The teleprinter may be located in the control room, relay room, computer room or other convenient and reasonably clean area such that the length of the wiring to the ATSI cabinet does not exceed 200 feet. For mounting dimensions, refer to the outline drawing supplied for the unit.

4.4 TELEPRINTER SHARING

Each ATSI system, by use of its unique identification character, is capable of sharing a single teleprinter with a second ATSI system. To incorporate this feature, simply add the following cross interconnections.

FROM UNIT "A"	TO UNIT "B"
+5VL	+5VR
+5VR	+5VL
TN SELECTED	TN BUSY
TN BUSY	TN SELECTED

Refer to the external connection diagram supplied with each system for terminal numbers.

4.5 CONNECTIONS

Make connections to the ATSI cabinet, detectors, and associated equipment as shown on the external connection diagram supplied for the system.

1. Grounding

WARNING

The National Electric Code requires electrical apparatus enclosures be solidly connected to building ground.

Connect a suitably grounded lead to the ground connection terminal provided in the ATSI enclosure.

2. Connections to Computer Input

If a computer or data logger is used, make connections to the computer input with shielded cable, as shown. Cut and tape the shield at the computer, and ground it at the ATSI cabinet end, as shown on the external connection diagram.

3. Vibration Detector Connections

Make connections to the vibration detectors with shielded cable (two wires within a shield, with plastic covering over the shield). For example, Belden No. 8422 cable (or

its equal) is suitable whenever the temperature is not over 165F. For situations in which higher temperatures will be encountered, use two-wire shielded cable of a type suitable for the temperature involved. Each detector is furnished with a plug and high-temperature cable assembly, usually 50 feet long. Generally any necessary extensions beyond the furnished cable may be made with ordinary plastic-covered shielded cable. However, should additional high-temperature cable be required, order by Cat. No. A5482265P2, specifying the length required.

The two-way resistance of the entire cable run for each vibration detector should not exceed 35 ohms. (The two-way resistance of the 50 feet of furnished cable will not exceed 2 ohms.) Avoid running the detector cable in strong electromagnetic fields or near conductors carrying heavy currents (50 amperes or more). If this cannot be avoided, run the cable in steel conduit or trays to provide electromagnetic shielding, being careful not to run any power wires in the same conduit or tray with the detector cables.

Dead-end the shields at the turbine end. Ground all shields at the cabinet end, as shown on the external connection diagram.

4. Connections to Position Detectors (Linear Variable Differential Transformers)

As shown on the external connection diagram, make the signal connection for each detector with shielded cable (two wires within a shield, with plastic covering over the shield). For example, Belden No. 8422 cable (or its equal) is suitable whenever the temperature is not over 165F. For situations in which higher temperatures will be encountered, use two-wire shielded cable of a type suitable for the temperature involved. Dead-end the shields at the turbine end and ground them at the cabinet end, as shown on the external connection diagram.

5. Eccentricity Detector

Connections to the eccentrcity detector are to be made with shielded cable (three wires within a shield, plastic covering over the shield). Belden No. 8403 cable (or its equal) is an example of a cable suitable wherever the temperature will not exceed 165F. For situations in which higher temperatures will be encountered, use three-wire shielded cable of a type suitable for the temperature involved.

Dead-end the shield at the turbine end and ground it at the cabinet end.

6. Differential Expansion and/or Rotor Expansion Detector

Connect each detector with shielded cable (three wires within a shield). The specifications for this cable are the same as for the eccentricity detector. Dead-end the shields at the turbine end and ground them at the cabinet end, as shown on the external connection diagram.

7. Vibration Phase Angle Connections

If vibration phase-angle measuring equipment is included in the system, make connections between the cabinet and the phase angle unit with multi-conductor shielded cable, as shown on the external connection diagram. Dead-end the shield at the phase angle unit, and ground it at the cabinet end, as shown.

Make connections to the vibration phase-angle reference detector on the turbine with two-wire shielded cable (same specifications as for the cable used for position detectors). This shield should be dead-ended at the turbine end and grounded at the cabinet.

8. Thermocouple Connections

Make connections between the thermocouples and the thermocouple amplifiers in the cabinet with thermocouple extension leads. They must be twisted pairs and shielded and of the same material as the thermocouples (chromel and alumel, iron and constantan, etc.). The polarity of commonly-used thermocouples is as follows:

Thermocouple Material	To be Connected to "+" Terminal
Chromel-Alumel	Chromel
Iron-Constantan	Iron
Chromel-Constantan	Chromel
Copper-Constantan	Copper

Thermocouples may be of the grounded or ungrounded type and should preferably be run in metal conduit or trays, and no AC power or control wires should be run in the same conduit or tray. All shields are to be connected at the cabinet only, as shown on the external connection diagram.

9. Grounding Circuit Common

On most systems, the circuit common (terminal no. 4 in the cabinet) is wired to some other equipment, usually electro-hydraulic control and there connected to ground. Refer to

the external connection diagram furnished with the system instructions. In general, if an outgoing wire is shown connecting from terminal no. 4 to another equipment, no jumper is to be used between terminals no. 3 and 4, as the circuit common will be grounded at the auxiliary-equipment location.

10. Alarm-Initiation Contacts Connected in Parallel

Special instructions apply whenever a number of normally-open relay contacts of dual-alarm circuit boards are connected in parallel. For details, refer to the INSTALLATION section of the publication for the dual alarm board.

11. Power Wiring

Connect the 115 VAC power to the cabinet with No. 14 AWG wire, using conventional wiring practice. Connect the neutral side of the line as shown on the external connection diagram (see notes on that diagram).

12. Video Coax Cable(s)

Properly terminated 75 ohm coax cables of the length specified at the time of the order are furnished for the video signal(s) that must be transmitted from the cabinet to the data monitor(s).

The cable length in increments of 25 feet (200 feet maximum) should be kept separated from all power and relay switching wires.

13. Trend Select Cable

Special multiple conductor twisted pair cable with overall shield for transmitting signals from the trend select unit to the cabinet is furnished with the equipment. The length will be the same as specified for the video cable(s) with the same precautions to be taken in the routing of the cable.

14. Teleprinter

As shown on the external connection diagram, make the connections to the teleprinter with shielded cable (#22 AWG minimum).

15. Other Wiring

For wiring other than that specifically discussed in this section, refer to the external connection diagram and use No. 18 AWG (or larger) and conventional wiring practices.

5.0 PRELIMINARY TESTS AND ADJUSTMENTS

The following preliminary tests and adjustments apply to the ATSI system as a whole. After all connections, per the external (customer) connection diagram, have been made and the tape recorder (shipped separately from the cabinet) mounted and connections made, the system is ready for preliminary tests and adjustments.

1. Energize the ATSI cabinet by switching on the two circuit breakers labeled UPS and LINE REFERENCE.

5.1 ANALOG SECTION

2. Turn the TRIP switch on the power supply ON. (Normally left on).

3. Turn the POWER switch on the power supply ON. (Normally left on).

4. The POWER pilot light should come on, and the time-delay should pick up (pick up of the relay should be heard).

5. Check the positive DC power supply voltage (the voltage at the red test jack on any of the circuit boards, with respect to a black test jack.) It should be +15.0 $\pm$ 0.3V.

6. Check the negative DC power supply voltage (the voltage at the green test jack on any of the circuit boards, with respect to a black jack). It should be -15.0 $\pm$ .3V.

7. Check for 118V $\pm$ 2 percent AC regulated voltage on terminals 5 and 6 of the power supply chassis, in the rear of the cabinet, with an accurate portable AC voltmeter. Note: All differential expansion, rotor expansion, shell expansion, position, and eccentricity detectors must be installed and connected to provide normal loading, in order to check the 118V regulated voltage.

8. Check for 50V $\pm$ 2 percent AC voltage on terminals 14 and 16 of the power supply, in the rear of the cabinet, with a portable AC voltmeter.

5.2 DIGITAL SECTION

1. Check that circuit breaker 1CB located in the rear of the digital bay is ON. This CB is provided for removing power from the digital section only, when changing out PCB's, testing, etc. and when it is desired to leave the analog section ON to retain its tripping capability.

   NOTE: 1CB also removes AC power from the data monitor(s). However, it does not remove signal inputs from the MUX board(s) or ±15V from the terminal boards in this bay.

2. Check that the five LED voltage indicators on the front of the power supply panel are ON.

3. Measure the four logic supply voltages at 5TB on logic rack #1 and verify that they are within the tolerance specified below:

	5TB (+)	5TB (-)	Tolerance
+15V	D	C	± 0.1V
-15V	C	E	± 0.1V
-9V	B	F	± 0.1V
+5V	H	J	± 0.1V

4. Check for +10V ± 0.1V on terminals M(+) and L(-) of 2TB on the logic power supply panel.

5. Load the tape deck with tape and check that it is ready for ON LINE operation. (Refer to separately listed instruction book covering the recorder.)

6. Set up the teleprinter as follows:

   a. Check that teleprinter has a sufficient supply of power.

   b. Set for all capital letters.

   c. Set single or double spacing as desired.

   d. Set print rate for 30 CPS or 120 CPS whichever is the higher rate selectable on tleprinter to be used*

   e. Turn teleprinter power switch ON.

   f. Place teleprinter on LOCAL and verify that the unit operates properly.

   g. Place teleprinter in STANDBY.

   *Whichever print rate is selected the setting of 1SW (BAUD rate switch) on the I/O board must agree. 1SW is normally set during factory test and should not require further adjustment. The setting of 1SW may be checked against the following chart:

BAUD \ 1SW (WT.)	1 (1)	2 (4)	3 (8)	4 (32)	5 (64)	6 (128)	7 (256)	8 (512)
110(10CPS)				X		X	X	X
300(30CPS	X		X		X			
1200(120CPS)	X	X			X			

X - contact closed

7. Check that the data monitor(s) operate locally (raster present). After check, turn intensity down until raster just disappears. LEAVE POWER SWITCH ON.

6.0 OPERATION

The equipment is basically automatic in its operation. Normally, monitoring of the CRT(s) and teleprinter printout plus performing routine maintenance of the recorder, cabinet, and teleprinter is all that is required. An automatic printout of a data log will be printed out each day at midnight but additional operator requested printouts is probably desirable. Besides examining the data logs, the printout should be examined for any warning messages and appropriate action taken.

For ordinary operation, keep the trip switch in the cabinet ON.

6.1 START-UP

After the preliminary test and adjustments have been performed, the system is ready for start-up.

1. Turn the teleprinter ON and place in STANDBY.

2. Check that the power switch on the monitor(s) is ON.

3. Check that the RUN/HOLD switch on the diagnostic readout board is in the RUN position and that the PROGRAM MODE switch is in the NORMAL position.

4. Turn power ON. (Switch on UPS and LINE REFERENCE circuit breakers.)

5. Check that the write enable light on the tape deck is ON. If it is not ON, check for a write ring on the tape reel.

6 Place the tape deck ON LINE. The top LED (Tape Ready) on the tape control board should be ON.

NOTE: Before tape deck will go ON LINE, it must be at the beginning of tape (BOT) marker (white reflective marker).

7. Set the calendar clock. Eight switches are provided on the clock readout board for this purpose. The switches allow a separate fast/slow setting of the year (two digit), day of the year, and hour (24 hours), minute and second, as well as a reset of the day and year counter. A maintained RUN/HOLD switch is also available which stops the clock while in the HOLD position, allowing an accurate setting of the seconds digit.

8. Push the lighted (red) PROGRAM START button located on the sidewall to the left of the logic modules. Pushing this button enables the 3 Hz interrupt, allowing the CPU to take over and start stepping throughs its program. After approximately six minutes of operation the first block of data will be recorded on tape with the tape having advanced approximately 2½ inches.

9. Determine and record BOT on the tape reel. Refer to teleprinter operation section for a method of determining BOT.

6.2 TREND SELECT UNIT

This unit allows the operator to select the desired trend and slope lines for displaying on the trend CRT. The front of the unit is divided into two functions. The left side contains alternate action lighted pushbuttons for selecting the desired trend channel (line). Pushing any one of these buttons will cause the button to light up and also cause a line (a series of dots spaced 45 seconds apart) to appear on the trend CRT which represents the past one hour of data for that parameter. Thereafter, this line will be updated every 45 seconds until the pushbutton is again pushed causing that line to disappear from the screen. Any number of these lines may be selected to appear on the screen at any one time.

The right side of the panel contains two rows of 11 push-buttons each, for requesting the microcomputer to generate slope lines in conjunction with their associated trend line. Each row of pushbuttons is associated with the trend line directly to the left of the dividing line as marked. One or both of the slope lines may be displayed at any one time, but may not be entered or cancelled simultaneously.

To select a slope line, you must first select its associated trend line. Then push the + or - slope, HI or LO scale, and one of the five slope rate pushbuttons associated with that trend line before pushing the ENT button. The ENT button will cause the buttons pushed to light up and the proper code to be entered into the microcomputer for processing. After a slight delay, the slope line will appear on the screen attached to the trend line at time zero and will thereafter be updated along the selected slope. Pushing the CAN button cancels the entry and causes the slope line to disappear from the screen. The END and CAN buttons are not lighted.

For those systems having only one CRT, the operator should first select the desired trend and/or slope line as described above, then depress the PRESS TO DISPLAY TREND button to view the lines on the CRT. A mechanical latch is provided to hold the button depressed but normally should not be used.

6.3 TELEPRINTER

The teleprinter is used for limited communications with the microcomputer. It will mostly be used for requesting a printout of a data log, although each night at midnight the microcomputer will automatically cause a log to be printed out as a check the system is functioning. This process is started when the microcomputer issues the command RECALL LAST BLOCK. This will cause the tape to first move forward, then reverse past the first block, then forward again to write the first log into post tape memory (POM). The microcomputer will then cause that log to be printed out on the teleprinter (the data could be up to 160 minutes old). This printout should be examined closely each day for any signs of abnormality.

The following is a summary of all the commands available to the operator. Each is covered in detail later in this section.

NOTE: Capital letters are required but slashes may be omitted.

1. Request data (RD) from tape -
   RD Δ * YR/MO/DAY/HR/MIN*

2. Print out next log (NL) in POM -
   NL Δ *

3. Print out present entry (PE) -
   PE Δ *

4. Cancel flashing bar (CB) -
   CB Δ * Bar #/0 or 1* 0 = enable flashing
   1 = cancel flashing

5. Enter beginning of tape (BOT) time & date -
   BOT Δ *YR/DAY OF YR/HR/MIN*

6. Enter last data entry (LDE) time & date -
   LDE Δ *YR/DAY OF YR/HR/MIN*

The following notations apply throughout this section.

* - returned character by microcomputer if data accepted

Δ - Enter assigned unique character. This character is required when entering all commands. Refer to the MI listed in the system TABLE OF CONTENTS for this character which is normally the turbine unit number.

NOTE: After the first asterisk has been returned by the microcomputer during the entry of any command having two asterisks, the operator must either complete the command or cancel it as the microcomputer locks out any further use of the teleprinter until this is done. Failure to do so prevents any messages or autologs from being printed out.

The REQUEST DATA command is used to search for data on tape to obtain a hard copy of a specific log. During the search, a block of 16 logs is captured in POM but only one log will be printed out. The procedure is as follows:

1. Place teleprinter ON LINE

   NOTE: The teleprinter must be on line during all communications with the microcomputer and should be returned to standby when completed.

2. Type in RD Δ.

3. Microcomputer will return an * indicating it is ready to receive further data.

4. Type in YR/MO/DAY/HR/MIN of desired log (if the requested time and date is not within the time frame of the tape, invalid will be printed out).

5. The microcomputer will again return an *, indicating it has accepted the command.

   EXAMPLE: RD3*76/07/02/20/22*

   After a time delay (3 minutes max.), the first log that the microcomputer detected that is nearest (equal to or greater than) the requested time will printed out.

   NOTE: After a search is started, the microcomputer will ignore all other commands from the teleprinter, until the tape is repositioned at POLDE. A 2 to 3 minute delay from the start of a log printout may be experienced.

If the nexy log is now desired, type in NL Δ. By repeating this procedure after each log is printed out obtains all logs within the captured block. For example, if the log requested turned out to be log 15 of a block, then typing in NLΔ will cause log 16 to be printed out. Another NLΔ will go back in time to log 1 of the same block, etc. If, however, a log from the next block is desired, request a new search by typing in a new time and date that falls within that block (longest = 80 minutes, shortest = 8 minutes except when a status or alarm condition occurs that requires an immediate log).

Under certain conditions, the first 2 or 3 logs of a block will not be examined for time and date. Therefore, all logs of a captured block may want to be examined by use of the NL command, if the log printed out is not within 5 minutes of the requested time. In lieu of the use of the NL command, a second search for the same time may be initiated.

Returning the teleprinter to standby while a log is being printed out returns a bad status signal to the microcomputer, which in turn cancels the command. A new command may then be entered immediately if the tape is at POLDE. This feature, however, is not available if current interface between teleprinter and microcomputer is being used.

The PRESENT ENTRY command is used to obtain data taken during the last scan of all inputs. This data may or may not correspond to a log that gets recorded on tape, as several scans are made between the taking of each log. The format is PEΔ.

The CANCEL BAR command allows the operator to cancel, or re-enable once cancelled, the flashing of any single or paired set of bars displayed on the Bar CRT. He may also re-enable the flashing of all bars as a group but, cancelling of all flashing bars as a group is not permitted. Paired bars are those bars with a common graticule between them instead of a graticule under each. They are also slightly closer together. Pairing is used only for those temperature parameters requiring a differential temperature limit.

The flashing of a bar on the CRT indicates the parameter's value has exceeded a preset limit. Flashing will continue until the parameters value decreases below its alarm set point. This command provides a way of cancelling the flashing of that bar once it has been noted by the operator, so that other parameters reaching their limits might be more apparent. Once cancelled, a bar will flash again anytime its value drops within its limit and then goes out of limits again. Also, all cancelling commands are automatically negated each day at midnight and must be re-entered if desired.

The bar CRT is such that up to 52 bars could be displayed (26 bars on the left half of the screen and 26 on the right). The bars are numbered starting with No. 1 on the upper left and No. 2 on the upper right, etc. Because each system layout is different, the bar's position number is used when entering the command. When cancelling the flashing of a paired set of bars, the lowest numbered bar of the pair must be used. Refer to the MI listed in the System TABLE OF CONTENTS for the bar numbers for your unit. Bar numbers correspond to the order in which data is transferred to the CRT shift register.

The command format is as follows:

CB Δ*bar#(2 digits)/0 OR 1 0 = enable flashing
1 = cancel flashing

EXAMPLE CBΔ*46/1*

Other than when searching an old tape, BOT & LDE times are automatically set in memory and requires no operator input. However, these times are required for accurate record keeping and when searching the tape once it has been removed.

The following format will be required when entering BOT or LDE and should be used when initially recording this data on the side of each reel. It is also recommended that the month, day and year be recorded along with the required format of the year and day.

Format: YR/DAY of YR (4 DIGITS)/HR/MIN Date: MO/DAY/YR

Example: BOT-77/0015/12/15 1/15/77
LDE-77/0100/08/30 4/10/77

It is recommended that the BOT time be determined, verified, and recorded on the reel as well as at the teleptinter for reference each time a new reel is installed.

The BOT time corresponds to the pushing of the PROGRAM START button or in the case of a new reel being loaded while the system is running, LDE of the old tape plus 1 second may be used as a more precise time is not required when searching old tapes. The true BOT, of course, could be up to 5 minutes past the LDE. If the exact time of the first log is desired for the record, the following procedure may be followed. After the first block has been transferred to tape, request a search for data just past startup time or LDE of the previous tape whichever the case may be. After a log has been printed out, using the NL-command determine the time for log #1 and record as BOT (Log #1 is the first log printed out having an earlier time than the previous NL printout).

LDE time should be determined before removing a reel of tape from the tape deck and may be determined as follows. Request a search for data in increments of 10 or 20 minutes going back from present time until the microcomputer accepts the date and starts a search. Once the search is completed, the NL- command may be used to locate log 16 of that blcok (log with latest time) which should be LDE. To verify that this is the true LDE, request a search for a time just past it. The response should be invalid indicating the last block was captured. If not, repeat the above using smaller increments of time.

If the turbine generator has been off line any time during the previous 80 minutes prior to a tape change, 5 minute increments should be used when requesting a search so as not to miss the last block.

From time to time, the microcomputer will automatically print out on the teleprinter one of the following messages for the indicated reasons.

1. INVALID (Reason-invalid data entered, invalid log captured in POM, or operator cancelled command).

2. T PARITY (Reason - microcomputer to PREM parity error).

3. END OF TAPE (Reason - end of tape marker detected - 25 ft. left).

Invalid will normally only be a response from an operator input but may be as a result of a bad block captured during autolog. The last two may occur at any time and therefore the printout should be examined at least daily for any of the messages and appropriate action taken.

6.4 CHANGING TAPE

A new reel of tape must be installed any time the reel is full, indicated by an alarm and the message END OF TAPE printed out on the teleprinter. Also, any time the tape deck is taken off line which includes the removal of power from the system the tape must be changed. It is recommended that a convenient time be scheduled for changing the tape before it reaches the end of tape marker which is 25 feet before the actual end. Fast logging is locked out beyond this marker.

When the tape is to be changed, a maximum of 45 minutes is available for removing the reel, cleaning the recorder, and getting the tape deck back on line without any danger of losing data.

To change the tape, the following procedure must be followed:

1. Through the teleprinter request a search and determine the time and date of the last data entry (LDE). This search must be completed even though the time of the LDE might be known. This will eliminate any possibility of a mark being left on the tape in an area that might cause a malfunction during subsequent searches of that tape.

2. Once LDE has been determined through a search and if the read indicator on the tape deck is still on, it is safe to take the tape deck off line.

   NOTE: If the write indicator is ON, this indicates a new block of data has been written on tape since the last search and step 1 must be redone to determine the new LDE.

3. Rewind and remove the tape.

4. Record LDE on the side of reel.

5. Clean recording head and tape path.

6. Mount and thread a new reel of tape. (To prevent skewed winding, the reel must be pushed firmly against its seat before locking in place.

7. Return tape deck to ON LINE status.

8. Determine BOT and record on the side of the reel.

NOTE: Refer to separate instruction book covering the tape recorder for instructions on how to operate the recorder using local controls, install tapes, and clean the read/write head.

6.5 SEARCHING OLD TAPES

The ATSI system is capable of searching recorded (old) tapes, that have previously been removed from the system, for the purpose of retrieving past data but only after it has been determined that the present turbine data is not desired.

To search an old tape, proceed as follows:

1. Place RUN/HOLD switch on the clock readout board in the HOLD position. This prevents logs, other than immediate logs, from being taken.

NOTE: Anytime 16 logs have been accumulated and the tape deck is at POLDE, the tape deck will start-up for a block transfer.

2. If the tape deck is still ON LINE, complete a search, determine LDE, rewind and remove the tape.

3. Load tape to be searched making sure the write enable ring has been removed and place recorder ON LINE.

CAUTION

LOSS OF DATA ON THE TAPE CAN RESULT IF THE WRITE ENABLE RING IS NOT REMOVED. MAKE SURE THE WRITE ENABLE LIGHT ON THE TAPE RECORDER IS OFF BEFORE PLACING THE RECORDER ON LINE.

4. Enter the BOT time and date as recorded on the reel to be searched using the following format:

BOT Δ *YR/DAY OF YEAR/HR/MIN*

5. Enter LDE in the same fashion as follows:

LDE Δ *YR/DAY OF YEAR/HR/MIN*

6. Proceed to search the tape as in a normal operation except the first date requested should be near BOT. This will save time as slow speed forward is used until the requested data is captured.

Upon completion of the search, rewind and remove the tape. Since the write enable ring has been removed no special precautions are required.

To return to normal system operation, without removing power, proceed as follows:

1. Press the PROGRAM STOP button.

2. Reset the calendar clock.

3. Reset the microprocessor and tape control circuits by momentarily lifting up on the handle of the RESET switch on each of these boards (located on the front bottom edge of these boards).

4. Load a new reel of tape and place the tape deck ON LINE.

5. Press the PROGRAM START button.

6.6 TIME CHANGE

When time changes from standard to daylight savings time or vice versa, the calendar clock on the ATSI system may be changed or not changed, at the customer's option. If the calendar clock is to be adjusted, logs should not be taken during the interval that the clock is being reset, which should normally take less than 2 minutes. The following procedure is recommended: If the time is to be turned ahead 1 hour, press the PROGRAM START button, set the calendar clock, then restart the program by pressing the program START button. However, if the clock is to be turned back 1 hour, a new reel of tape must be installed. First, to prevent losing data in PREM, the operator must wait until a block of data has just been transferred to tape, indicated by the tape having advanced approximately 2½ inches. Note the time, for use as LDE of that reel of tape. Next, request a search. Once complete, take the tape deck OFF LINE and press the PROGRAM STOP button. Next, reset the calendar clock by advancing the hr., min., and seconds into the next day. Once this is set, reset the day before pressing the PROGRAM START button. Once the program is running, the tape may be rewound, a new reel installed, and the tape deck placed back ON LINE.

6.7 DIFFERENTIAL EXPANSION AND ROTOR EXPANSION

Equipment installed and connected in accordance with instructions will read forward clearance; that is, an increasing reading denotes increasing forward clearances at the point on the turbine where the detectors are located.

6.8 VIBRATION

While starting and stopping the turbine, vibration points will be displayed and recorded at or near zero at turbine speeds below 400 rpm, since at these low speeds the vibration frequency is too low to actuate the equipment. Vibration readings should be considered approximate from 400 to 780 rpm, and accurate at speeds above 780 rpm.

In case a vibration detector or detector cable becomes open, the bar graph will indicate zero mils. The indicating milliammeter (if one is used) will indicate zero ma, which is an abnormal reading, since 1-ma output represents zero vibration.

6.9 STARTING AND STOPPING

In general, it is recommended that the Advanced Turbine Supervisory Instruments be left on at all times. However, there may be occasions when it is necessary to shut down the instrument system temporarily, for servicing or other reasons.

NOTE: Before removing power from the digital section, you must first complete a search routine so as not to leave a mark on tape, that can cause a malfunction during subsequent searches of that tape.

To turn off all equipment in the cabinet, turn off the UPS and LINE REFERENCE circuit breakers. To turn power off the analog section only, turn the power switch on the power supply panel off. To turn power off the digital section only, turn circuit breaker 1CB off.

6.10 POWER FAILURE

The ATSI system requires an uninterruptible source of AC power, that will continue to operate all equipment in the system except the teleprinter for at least 30 minutes after loss of the standard AC power.

If the AC power has not been restored before the UPS reaches the end of its capacity, then the system will stop operating and any data that was left in RAM will be lost. When power is again restored, the system will automatically power back up but the start-up sequence will have to be followed before normal recording and displaying operations will be restored. However, the analog section will return to normal operation. It will require only about 30 seconds for the vibration readings (meter if used) to settle down to their normal levels and the trip circuit will be automatically disabled until about 45 seconds after restoration of power, at which time its normal tripping capability will be restored.

CAUTION

DO NOT ATTEMPT TO ADJUST THE VIBRATION ALARM/TRIP SET-POINT DIALS WHILE THE TURBINE IS RUNNING, AS THIS MIGHT TRIP THE TURBINE.

7.0 PRINCIPLES OF OPERATION

7.1 ANALOG SECTION

Fig. 8 is a block diagram of the analog section of the ATSI system. However, it should be understood that this diagram is illustrative only, and does not purport to represent any particular system in detail. The user's system may have more channels of a given measurement than shown, some of the channels showm may be omitted or additional measurement channels, not shown, may be added.

The detectors (except the speed detectors) generally provide AC voltages which are proportional to the quantity being measured. These AC voltages are then sent to the various amplifier boards in the cabinet, where the following necessary operations are performed.

The vibration-detector voltage is integrated, amplified, passed to a peak-reading circuit, then to a DC power amplifier. The position-detector voltage is sent through a phase-sensitive rectifier (demodulator), to provide a DC voltage output, which is then amplified. The speed detector output pulses are converted to a voltage by use of a frequency to voltage converter, and then amplified. The eccentricity-detector voltage is amplified, rectified by a phase-sensitive rectifier and filtered, and the resultant signal then operates a peak-to-peak circuit, followed by DC amplification. The vibration phase-angle measurement channel has a special mode of operation (see instruction publication for that measurement channel). The expansion detector voltages are sent through phase-sensitive rectifiers, which produce a DC voltage output, and this voltage is then amplified. In each case, the basic output of the amplifier is 2 to 10 volts DC, which is modified to provide 0.2 to 1.0 ma for chart recorders (if used), 1 to 5 ma for indicators, and 8 to 40 mv for a computer input.

The acceleration channel operates by differentiating the 2 to 10 volt output of the speed amplifier. Only one output is provided by the acceleration channel, -5 to 0 to +5 ma center scale.

For a more detailed description of the principles of operation of any particular measurement channel, refer to the instruction publication for that measurement channel.

7.2 DIGITAL SECTION

Fig. 9 is an overall block diagram of the ATSI system with the inputs and the microcomputer subsystem highlighted. The detectors and associated analog amplifiers referenced on the input are covered in the analog section.

Most status inputs are contacts. Appropriate resistor networks are used to produce at the MUX input a voltage level that interprets into good or bad status. (Refer to the system elementary diagram for specific circuits.)

Throughout the following circuit discussions refer to the PW board elementary diagrams as listed in the TABLE OF CONTENTS of the system instructions.

7.2.1 Signal Conditioning and Digitizing

Thermocouple Amplifier PWB

Each thermocouple amplifier PWB contains eight segregated channels which amplify low level signals from thermocouple junctions to higher levels prior to multiplexing and data conversion. Each channel provides a DC gain of 150 and filtering to reduce common mode and differential mode noise. Potentiometer 1P provides the means of adjusting the gain of the stage and 2P allows for adjustment of offset voltages of the amplifier. Each channel is adjusted at the factory and should require no further adjustments.

Each channel also has a failed thermocouple detection feature which causes the output to go to negative saturation when a thermocouple opens. This is accomplished by 1R and 3R which provide a bias current for the input of op amp 1A, and 2R and 4R which provide an offset voltage which drives 1A into negative saturation if the thermocouple opens. During normal operation, the low impedance of the thermocouple shunts out the effect of this bias.

Multiplexer (MUX) PWB

All input signals to the digital section are via a multiplexer (MUX) (Block A of Fig. 9). The number of inputs per system determines if one or two boards are required.

Each MUX board selects two of 64 inputs and applies them to the two outputs. The desired pair of inputs is determined by five bits of address information contained in HA0 through HA4. The first two bits, HA0 and HA1 are applied to each of the eight analog multiplexers and selects two of the eight possible channels from each multiplexer. The last three bits

of the address selects the appropriate multiplexer and applies an inhibit signal to the other seven devices on the board.

The board also contains protection circuitry for the multiplexers and +12V, -12V, and -2.1V power supplies. The +12V and -12V supplies generated on this board from the +15V and -15V supplies are used on this board, as well as being fed to the backplane board for use by other boards. If more than one MUX board is used, these supplies would automatically be connected in parallel through the backplane wiring.

A/D PWB

(Refer to blocks D, E, G & H of Fig. 9 and the board elementary). The primary function of this board is to select one of the possible eight analog input signals received from the mux board(s), amplify it, apply either a 2V fixed bias or cold junction compensation, and convert the resulting analog voltage to an eight bit digital word.

To accomplish this function, two multiplexers (MUX), five operational amplifiers, two multiplying D/A converters (MDAC), three E-PROMS, and an A/D converter are used.

Address inputs HA5, A6 & A10 selects one of the eight analog inputs applied to MUX E7 and applies it to the non-inverting input of amplifier E6. This amplifier performs the function of a unity gain buffer and provides a high input impedance for the analog input. The signal is then amplified via 1MDAC and amplifier E3, by a factor ranging from 0 to 5 as determined by the program in PROMS C5 & C6. The signal is then applied to summing amplifier E2 where, either a fixed bias or cold junction compensation is applied before being converted to an 8 bit digital word by the A/D converter.

Cold junction compensation is provided by the use of a temperature sensitive resistor (RT) with a controlled linear temperature coefficient, which is mounted inside the thermocouple amplifier box. Resistor RT plus amplifier D1 and two precision current sources (1VR for resistor RT and associated resistors and 2VR for zener diode 1BD) comprise the circuit, with its output appearing at TP6. The signal is then selectively applied via MUX C2, 2MDAC, and amplifier E1 to summing amplifier E2. PROMS C3 and C6 control the gain of amplifier E1.

Input voltages from the analog amplifiers have a 2-10V range and require a fixed bias to result in 0 volts out for minimum input. This is accomplished with a -1V fixed voltage at the wiper of 3P where it is selectively applied in the same manner as cold junction compensation.

Address inputs A7 thru A9 selects the address applied to PROMS C3, C5 & C6. The PROMS program then sets the address applied to the MDAC which sets the gain of operational amplifiers E1 and E3. PROM C6 also addresses MUX C2 which controls the application of the fixed bias or cold junction compensation.

Through the use of the PROMS and MDACS, eight different gains are obtained. Table 1 on the board elementary list the gains used for each type signal.

Once converted to a digital word by the A/D converter, the resulting word is applied to the appropriate memory data bus pins by tri-state buffers. GATED EMUX initiates the conversion and EMUX enables the appropriate tri-state buffers.

There is also a data routing function on this board which applies either the digitized analog signals or the calendar clock data information to the memory data bus. The clock data is applied via tri-state buffers, to the memory data bus when ECL becomes a logic "0".

The calibration of this board is done at the factory and should require no further adjustments. If adjustments are required, return the board to the factory.

7.2.2 Microcomputer Subsystem

The overall operation of the ATSI system is directed and supervised by a microcomputer subsystem consisting of three printed circuit boards each of which provides one of the basic functions of the subsystem. These boards are identified as microprocessor, I/O, and PROM boards. The general way in which the component circuits of the microcomputer subsystem act to control the overall system will be discussed with reference to the block diagram of Fig. 9 and in terms of the individual boards.

The three boards of the subsystem correspond approximately (but not exactly) to the three basic elements of any computer system whether micro, mini, or maxi and these are: central processor (CPU), memory and input/output control. The main exceptions are that

read/write memory (RAM) is located on the microprocessor board and that there are a number of eight-bit selector switches on the PROM board.

Microprocessor Board

This board contains the Intel 8080 eight-bit microprocessor which is capable of directly addressing 65,536 bytes of memory. It, also, has input/output instructions that permit an additional 256 locations to be addressed as input ports and another 256 locations as output ports. The three most significant bits of the 16-bit address bus are decoded on this board into eight lines, each of which is used to enable an 8K block of the memory space. Of the eight blocks thus defined, three are assigned to PROM, one to RAM, one to the realtime clock signals, one to the input signal multiplexer, one to the byte switches used for setting limit values and one spare block.

Also contained on this board is a 10 MHz crystal controlled oscillator from which are derived the phase 1 and phase 2 clocks at 1.67 MHz for the microprocessor and also phase 1 and phase 2 clock at the same frequency for use at TTL levels.

There is a power-on reset circuit as well as a manual reset circuit controlled by a momentary-contact switch on the front edge of the board. The reset function is used to reset not only the microprocessor itself but also the DMA (Direct Memory Address) counters, the interrupt circuits and the output port latches.

Included on this board is 1K bytes of RAM made up of eight 2102 1K X 1 RAM devices.

There is a DMA bus over which a block of RAM data with parity is sent at 1/3 second intervals at high speed (417 KHz) to the CRT master board. Another block of data is sent over this bus at 0.5 minute (fast log rate) or 5 minute (normal log rate) intervals at 80 KHz to the PREM board in the tape section. The address counters for the DMA operations are located on this board. DMA operations are initiated when the program places the necessary signals on output port 0 located on the I/O board. The signals from this port are received on the microprocessor board at pin 78 as a HOLD REQUEST signal and at pin 81 as a START CRT DMA signal. The HOLD REQUEST signal arrives first and acts to cause the microprocessor to go into a HOLD state. The START CRT DMA signal arrives next in the case of DMA transmissions to the CRT section. When DMA data is to be sent to the PREM board, the second signal, $\overline{\text{TDTP}}$, is received from the PREM control board.

The second signal in each case starts the DMA counters and starts the transmission of data over the DMA bus. At the end of the transmission, a terminal count signal from the counters causes the microprocessor to leave the HOLD state and to continue execution of the program.

A timer, started at the time the HOLD signal arrives on pin 78, is used to reset the HOLD flip-flop H3, after a fixed time delay and take the microprocessor out of the HOLD state in the event of a fault in the DMA circuit.

The address and data buses are buffered before they leave this board. Address line HA0 through HA5 are provided at 12 volt levels for use on the MUX and A/D boards.

Circuitry is provided to add a wait state in every instruction cycle to provide extra time margin in memory read operations.

There is a status circuit latch from whose outputs are derived the signals for controlling memory and I/O read/write operations. Outputs of the status latch are also wired to LED's mounted along the front edge of the microprocessor board.

I/O Board

This board contains seven output ports and six input ports through which the microprocessor communicates to other parts of the system via the I/O bus. Input port 1 is used to receive from the POM board data that results from a search operation. Input port 2 is used to receive other information from the tape section and input port 4 is used mainly to receive slope information from the Trend Select Unit.

Output port 1 is used to send commands to the tape section and output port 0 is used for sending DMA information to the tape and CRT Sections. Output ports 3 and 6 are connected to LED indicators mounted on the front edge of the board to supply interrupt status and other diagnostic information.

Also mounted on this board is a universal asynchronous receiver/transmitter (UART) circuit for simplifying communications with the teleprinter. Output words to go to the teleprinter are sent from the microprocessor board to output port 4 where it is latched and passed in parallel to the UART via a strobe pulse transmitted via PD1 on output port 5. The UART sends the word in serial form out its pin 25 through a buffer and out board pin 84 to the teleprinter.

Incoming words in serial form from the teleprinter are received on board pin 83, go into UART pin 20 and are assembled as words on UART pins 5 through 12. The assembled words are read in parallel into the microprocessor via the I/O bus by means of a signal on RDE, pin 4 (input port 0).

A selectable clock rate generator for the UART is provided on the I/O board. Clock rates corresponding to different baud rates are selectable by means of an eight-bit selector switch on the board in accordance with instructions given on the I/O board elementary diagram.

A detailed explanation of the assignments of all pins on all I/O ports is given on the system elementary diagram.

Also located on this board is most of the interrupt circuitry external to the microprocessor itself. The identy of the interrupt to be serviced at any particular time is established on the I/O board but the latch for holding the restart instruction to be placed on the microprocessor data bus at the appropriate time is on the microprocessor board.

There are provisions for latching in interrupt signals until they are serviced and for determining the priority of interrupts which arrive at the same time. All interrupts are gated so that they can be selectively disabled by signals from the microprocessor via the I/O bus.

PROM Board

The PROM board has 24 pin sockets for EPROMS arranged on the board in ascending address order from left to right and top to bottom starting with the lowest address and ending with the highest address.

Also on this board are a number of position selector switches used as a means for setting alarm level values specified for certain measured quantities. They are addressed as memory.

PROMS are used in the ATSI system to hold the application program. The wiring on the backplane allows the microprocessor to address the PROMS as well as enable them in blocks of 8K bytes.

7.2.3 Tape Memory Subsystem

Consisting of:

PREM (Pretape Memory)

PREM Control

Tape Control

POM Control (Post Tape Memory Control)

POM (Post Tape Memory)

and Tape Deck

Clock Generator

Several clock pulses are used in the Memory System (MS) that are generated by the clock generator circuit on the POM control board.

The 80 KHz clock signal is generated and assigned three different names which are used when data is received from the microcomputer. To distinguish these signals from the normal clock signals (CP) used in the MS we call them CPU-shift, CPU-S5, and and Tape DMA clock. The three (3) are distributed as follows:

1. CPU-shift - 12V level signal used on the PREM Control board.

2. CPU-S5 - 5V level signal used on the PREM board.

3. Tape DMA clock - 5V level signal used on the microprocessor board.

Memory System Clock

Several clock rates are used throughout the Memory System for purposes of synchronization, timing and other functions.

The basic clock rate is CP equal to 320 KHz. Several subharmonic rates are used. The number before CP indicates the divisor by which CP must be divided to obtain the pulse rate in question. 32CP corresponds, therefore, to 10 KHz, the rate at which data is written onto the magnetic tape.

A special shift pulse is the SR SHIFT. This signal produces a pulse required by the dynamic shift register and compatible with the tape deck. This pulse of 6.3 us is produced at the end of each 32CP cycle and advances the shift register 1 bit. The data is read into the tape deck at the positive going edge of the 32CP or midway between SR shift pulses.

Pretape Memory (PREM)

Data Handling

Data is moved from the microcomputer to the PREM in log form. Each log contains up to 111 words of log identification, time and date, turbine number, etc. and, of course, data from the various sensors of the turbine. Each word is an array of 8 parallel bits or binary digits.

The PREM receives these logs one at a time, normally, once every 30 seconds or every 5 minutes depending on the alarm status but under certain conditions an immediate log is taken. The PREM assembles these logs in a long string of words until 16 logs or a data block is complete.

This block is now transferred to the magnetic tape deck just the way it was stored with log #1 first and log #16 last. Special code bits are added to the tape to verify the proper data transfer. These code bits are read right after the writing process and only after correct codes have been recognized will the PREM be permitted to erase those logs that have been transferred. A second block transfer will be attempted if incorrect codes were read.

A block transfer is only possible if the tape is positioned at POLDE (point of last data entry). Should the tape be at any other point, e.g., during a search, the block transfer has to wait. Logs, coming in during this time, are still accepted by the PREM

The PREM has an overflow capability of 8 logs of maximum length beyond the 16 logs that form the next block. That block will be transferred as soon as POLDE is reached again and the previous tape deck operation has ended. Once the overflow limit is reached, software prevents any further logs from being taken until POLDE is reached and the next block is transferred to tape.

Data Format and Codes

The output of the PREM determines the data format that is written on the tape. The data format used is IBM compatible with the following detailed specifications.

Recording Method	NRZI (Non Return to Zero Invert) Logic "1"s are indicated by a flux inversion. No flux inversions (at defined locations) are logic "0"s.

Recording Density	800 bits per inch
Tape Speed	12.5 inches per second
Number of Parallel Tracks	9 (8 data tracks, 1 parity track)
Block Length	426 to 2048 words 0.5 to 2.5 in. of tape
Length of Interblock Gap (IBG or IRG)	0.6 in.
Codes:	
Vertical Parity (VP)	Odd; located on 9th track
Cyclic Redundancy Code Character (CRCC)	Located 4 spaces after the last data word
Longitudinal Redundancy Code Character (LRCC)	Located 4 spaces after the CRCC word

The vertical parity code is essentially a word parity bit. The VP is a logic 1 if the word has an even number of 1's and vice versa. Even the CRCC and LRCC have odd parity in this application, which involves only blocks of even number of words. Any gap, like the IRG, is formed by a RESET flux on all channels which correspond to the magnetization of a virgin tape. The LRCC re-establishes that RESET flux after writing a block of data. The LRCC uses even parity. The CRCC is a complex code pattern that, in a simple way could be described as a diagonal parity code.

Inserting and testing for the presence of correct codes assures a very high probability of correct data storage.

Pretape Memory (PREM)

The PREM stores temporarily data which was presented in the form of individual logs. Each log contains a lead word that identifies the logs position in a data block. Lead words would be 1 through 16 (254 through 239). The lead word is followed by up to 110 data words. Last, each log contains an end word (255). Logs will form one continuous string when stored in the PREM.

Data is entered through the terminals DDO through DDP. Inverters convert the negative logic of the system to

a positive logic. Proper parity of the incoming data is verified by the parity checker (74180). Actual error signal generation is done in the PREM Control circuit.

Memory circuits consist of dynamic shift registers. Data is either recirculated (SR CONTROL 2 low; 1 high) erased (both SR CONTROLS high) or written which means transferred from the microcomputer (SR CONTROL 2 high; 1 low). The data is kept moving by a special 10 KHz clock signal (SR-SHIFT 5) generated in the clock generator. TAPE SHIFT is used for data transfer to the tape deck. Only during data transfer from the microcomputer to the PREM are the shift registers operated using the CPU-S5 signal (80 KHz).

Data is transferred to the tape deck after 16 logs are accumulated if the tape deck is not busy with something else. The memory is actually capable of storing 24 logs of max. length, 8 of which represent an overflow capacity if the block transfer (the first 16 logs) has to be delayed with more logs coming in.

The block transfer is initiated by the PREM Control circuit. Proper data train and log identification is done by the log label. The 3 most significant bits are combined to indicate a log mark (when asserted). LL1 through LL5 indicate the lead word or log number.

All register outputs are connected to outputs that connect to the tape deck's write inputs. Vertical parity is regenerated to satisfy the corresponding tape deck input. The CRCC generator accepts the same data that is shifted into the tape deck. After one more shift with its input kept low, the output switch flip-flop removes the output connection from the shift registers and connects them to the CRCC generator. A reset signal re-establishes the starting condition of the code generator at the end of the block transfer operation.

The PREM uses a 5V positive logic level that mixes T^2L - C-MOS - RT logic and N-MOS LSI and MSI circuits on one card. Signals leaving the board are 5V or 12V negative logic levels corresponding to the tape deck or the interface circuits.

Pretape Memory Control (PREM-Control)

The PREM Control operates the PREM through its main functions.

1. Recirculate data at SR-SHIFT (10 KHz) speeds and accumulate data at tape DMA speeds (80 KHz).

2. After a request for log transfer is received: Switch from SR-SHIFT to CPU-SHIFT. Synchronize the microcomputer with data stream. Add new log at end of datastream. Check for parity. After transfer, go back to SR-SHIFT.

3. After 16 or more logs are in memory, synchronize tape deck start up with data stream. Transfer data. Add CRCC code. Add LRCC code. Check for correct read after write. If OK, erase first 16 logs in PREM. If not OK, repeat whole sequence. Report to the microcomputer and erase.

SR-CPU - Shift Switch

The PREM has to be switched from the SR-SHIFT to CPU-SHIFT if a log is to be transferred from the microcomputer to the PREM.

The SR-SHIFT is a negative going 7.0 us wide pulse repeated at 10 KHz rate. The CPU-SHIFT signal is a 80 KHz square wave. A switch over will be done when the SR-SHIFT signal is high and the CPU-SHIFT signal goes high. Certain other conditions have to be met in order to obtain a switchover without a glitch that could cause pick up or loss of individual bits in the memory. The shift speed of the PREM is returned to the SR-SHIFT as soon as the log transfer is completed. Again the shiftover occurs when the SR-SHIFT signal is high and the CPU-SHIFT goes high.

Transfer Log

This is the function that synchronizes the log transfer with the data stream in the PREM so that the next log is attached to the tail end of the data stream. Two cases have to be considered. First, if no data is in thePREM (which is recognized by a timer; a 14 stage counter and a 10 KHz pulse), the go ahead can be given after one revolution of the PREM during which no data train was detected. A time of 210 MS was selected. The leading edge of the signal TRANSFER DATA INTO PREM must also be synchronized with CPU-SHIFT.

The second case involves finding the trailing bit (end word) of the data train in the PREM and generating the signal TRANSFER DATA INTO PREM just before the next shift occurs. The DATASTREAM signal which defines the leading and trailing edge, is used for that purpose. The new log is then attached to the trailing end of the stored data. The microcomputer removes the signal

CPU REQUEST LOG TRANSFER right after the last (end) word of the new log. The PREM Control changes the 2 SR-CONTROL signals from the writing to the re-circulate condition.

	SR-Control 1	SR-Control 2
Erase	12V	12V
Write	0V	12V
Recirculate	12V	0V

Parity Error of Microcomputer Data

Every word transferred from the microcomputer to the PREM is checked for correct odd parity. Any parity error will set the output PARITY ERROR. This signal is then sustained until the end of the signal CPU REQUEST LOG TRANSFER.

Data Stream Signal Generator

This function detects the presence of the data stream, using lead words and end words of the logs. Its basic function is to determine the tail end of the data stream for attaching additional logs and for determining the length of the 16 logs that form the block that has to be transferred to the tape deck.

The leading edge of the data stream is found using a stream of "0" data words which represent the unused portion of the PREM (the data stream is of course flying around in the dynamic shift register). The leading edge is detected if the zero stream ends and a log mark (first lead word) appears. The tail end of the data stream is detected if an end word is followed by a "0" data word.

Start Tape Timer

This function synchronizes the movement of the data stream with that of the tape deck. The timing process is accomplished by counting the shift pulses from the front end of the data stream passing the end of the shift register. The counter is preset to a number that is smaller than the total length of the shift register. The tape deck is therefore started before the data stream reaches the end of the shift register. The count will be adjusted in the factory so that a gap of 0.6 in. exists between data blocks.

Block Transfer Logic

This logic senses the presence of 16 logs in the PREM. The block transfer is initiated as soon as conditions permit. The data is then removed from the PREM after the read after write check verified a correct transfer.

The START 16TH LOG flip-flop (FF) generates a signal at the time the 16th log passes the end of the PREM. This signal indicates to the tape control the need for a block transfer. The signal is repeated at each passage of the log stream. The tape control will reserve the tape deck for this operation. Start the tape deck and verify to the PREM control BLOCK TRANSFER IN PROGRESS.

With the tape deck running at full speed, the data will arrive at the output of the PREM 0.6 in. after the end of the previous block. The signal TRANSFER DATA OUT OF PREM (TDOP) is then issued to apply the write signals that shifts the information onto the tape.

After the last word of the 16th log was shifted onto the tape the signal TDOP is turned off. The code space counter is started. The signal ADD CRCC is issued 4 spaces after last data word. The signal ADD LRCC is then added after another 4 spaces.

With the block transfer in progress and the 16th log being transferred, a FF (look for code OK before next stream) is set. The NO CODE ERROR signal is received from the POM Control about 12.8 ms after the last (end) word of the block was written. This sets a FF (block transfer OK).

Flip-flop ERASE DATA is now activated starting with the first word of the next arriving data stream. The input gates of the PREM are simply set to insert "0"s into the register instead of the block which was transferred to the tape. Any log that was in the overflow is now recirculating as the first part of the next block.

If the NO CODE ERROR signal does not arrive during the time interval end of the 16th block - start of next data stream, the code error FF comes on. This resets the first FF (look for code OK before next data train). The whole block transfer is now repeated. The data of the block is then erased even if another read after write error occurred.

Last PREM Log Frame

The last log in the PREM has to be accessible to the microcomputer. A comparison of a buffer storage set to the last lead word in the PREM and the actual lead word read is used to find the beginning of the last log frame. The following end word is used to indicate the end.

The DATASTREAM signal is used to clock a FF. First the FF is in the even state. A buffer memory is set to every lead word value (5 least significant bits only). This means that the last lead word will remain in this memory during the next data circulation while the FF is in the odd state. A comparison circuit looks at the buffer memory and the log label numbers (from the PREM). If coincidence exists the last lead word or front end of the last log was found and a FF is set. The next end word indicates the end of the log and the FF is reset.

The LAST PREM LOG FRAME signal is used by the POM Control to channel the actual log data to the microcomputer.

NOTE: Two(2) logs are marked if the PREM overflow is used. However, the call for a last PREM log should not coincide with use of the overflow. The overflow should only be used during a search.

Tape Control

Tape Ready

TAPE READY is a signal which tells the microcomputer that the tape deck and the memory system are ready to listen to its commands. This does not cover any activity between the PREM and the microcomputer. Log transfers can continue even if the tape deck is turned off.

The TAPE READY signal will come on after a loading operation has been completed, the tape is at the BOT mark, and when the ON LINE pushbutton (on tape deck) is pushed, the tape will then advance 5 in. which is one visual confirmation of the TAPE READY condition.

The TAPE READY signal will go off every time the tape deck is taken off line (ON LINE on tape deck goes off). This will then be interpreted by the tape deck as a change of tape. Any memory as to tape versus date and time of data will then be erased by the microcomputer and a new tape must be used.

The normal tape change sequence is as follows:

1. Manual Reset (takes tape off-line)
2. Manual Rewind (rewinds tape to load point)
3. Manual Rewind (rewinds tape completely)
4. Remove old tape
5. Clean tape path
6. Install new tape
7. Manual load (advances tape to load point)
8. Manual On-Line (advances tape 5 in.) Ready for microcomputer commands.

5 or 10 in. Advance Detector

This function serves 2 purposes. The first use is the initial 5 in. advance right after putting the tape deck on line. Besides being an IBM format requirement, this advance moves the BOT tab past the magnetic head. The tab might interfere with the proper writing and reading of data.

The second use is to find the POINT OF LAST DATA ENTRY (POLDE). Every time the tape deck is started in slow forward (exceptions are: on-line-load, in write mode), a timer is started to look for a 10 in. interval of tape without blocks of data. Each time a block of data is recognized, the timer is reset. When the timer finally counts out, 10 inches of empty tape went by that are just beyond POLDE. The tape control uses this condition to interrupt the slow forward motion and initiates a RECALL LAST BLOCK. operation.

A REQUEST SEARCH signal from the microcomputer, that was executed and completed in the above manner, must first be negated before a new search request will be honored. The signal is negated when the tape is repositioned at POLDE.

Tape Deck Deceleration Timer

Deceleration of the tape deck starts after the motion command signal is removed from the tape deck. A timer is used to prevent another command from interfering with the proper deceleration of the tape deck drive.

Any motion command will set a memory (FF). Returning the motion command to the off state starts the counter for the short interval (slow speed) or long interval (high speed).

The short interval is 0.0512 second (512 counts at 10 KHz) and the long interval is 0.8192 sec (8192 counts at 10 KHz). After timing out, the signal TD IN MOTION is removed and the next control signal may be applied.

Time-Speed Product Counter

This counter provides an indication, at the control cabinet, via the CHANGE TAPE LED on the tape control board, if the product of time of operation times tape speed exceeds a limit. The present limit is set for

$$(T_F \times V_F) + (T_S \times V_S) = 400 \text{ to } 600 \text{ hours in/sec.}$$

Any fast tape motion will apply 512CP signals to the counter. The counter would reach its limits after 3.72 hours. A slow tape motion advances the counter at 4096 CP and signals the limit after 30 hours. This function is for indication only and indicates that the end-of-tape region may be near. The operator can ignore it if he wants to. No reset is provided for the operator to cancel this remainder but is automatically reset during a tape loading procedure.

End of Tape

Built into the tape deck is a memory that indicates whenever the tape is in the end-of-tape (EOT) region. The tape control output - END OF TAPE - is always affected when the tape deck is in this region. The microcomputer will issue an alarm to the operator and a LED on the tape control PWB will light indicating the end-of-tape condition. This means, of course, that the tape must be exchanged as soon as possible. Once the end of tape indication is given, a mere 20 feet of tape is left. Only slow operation is possible in the EOT region to avoid an accidental run to the real tape end and loss of tension.

The Function Scanner

Several different functions can be distinguished that have to be executed at various times by the tape deck.

These are in descending order of importance:

Block Transfer

On-line Rewind

Search

Recall Last Block

The request for a block transfer indicates that the (limited) PREM contains a data block and any future logs would have to go into the overflow memory. A request for block transfer must always be honored first or as soon as possible. The signal from the PREM which requests the block transfer is of very short duration (it is the equivalent of the lead word of the 16th log). This signal is then repeated once for every revolution of the PREM until the request has been honored. This means that the function scanner must test for the request for at least 307.2 ms. A 400 ms timer is used to accomplish this. Details of the block transfer are discussed later. The on-line rewind would normally only be used at the beginning of a search for data contained close to the beginning of the tape. The microcomputer issues only a short signal that is removed as soon as the signal SEARCH IN PROGRESS is issued. The function scanner will go on and stop at the next position SEARCH upon completion of the on-line rewind.

The next function of importance is the search for a previously stored block of data. In this mode, the tape deck is under full control of the microcomputer. This mode is started by the CPU REQUEST SEARCH signal. Once the function scanner has latched on and issued the response SEARCH IN PROGRESS, the microcomputer can take over control.

Every search must be finished by running the tape in slow forward past POLDE so that the tape can be re-positioned at the end of the data blocks on the tape. Only then is a search complete and the deck ready for the next command. Details of the search are discussed later.

The last function the function scanner can select is the RECALL LAST BLOCK operation. Every search is automatically terminated by this function. If the tape goes (slowly) by POLDE, a timer starts and limits the forward motion to 10". The function scanner advances then to RECALL LAST BLOCK. The tape deck will reverse until it passes the last block. Forward motion is then applied again. The end of the last block is then used to turn off the tape deck.

The tape is now positioned at the end of the last block ready for the next block transfer.

The RECALL LAST BLOCK can also be used to simply recall the last block. The microcomputer issues in that case the corresponding signal. The signal must be removed as soon as the response SEARCH IN PROGRESS

is issued. The tape deck will write forward 2 inches, move back one block, advance one block and stop. The last block will then be stored in the POM.

Block Transfer Details

The block transfer request originates in the PREM control. The lead word of the 16th log triggers the signal REQUEST BLOCK TRANSFER. Coincidence of that signal and the test signal from the function scanner, sets the flip-flop BLOCK TRANSFER IN PROGRESS. This stops the function scanner and applies the signal SWS (set write status) to the tape deck. This simply means that the next command to the tape deck (TD) will be a write instruction. Synchronization of the actual tape startup and the data block, which is rotating in the PREM, is done by the signal START TD FOR BLOCK TRANSFER. This signal precedes the appearance of the first word of the block at the PREM output terminals by about 36 ms. Timing is done so that the tape deck is up to full synchronous speed and an IRG of 0.6 inch is obtained at the time the first word is written.

Data is now written on the tape as it appears at the output of the memory. It remembered that the data is circulating in the PREM the output of which is connected back to the input. The tape motion and the circulation speed in the PREM are synchronized so that the bit density on the tape equals 800 bits per inch.

Data transfer from the PREM to the tape deck ends with the end word of the 16th log by turning off the signal TRANSFER DATA OUT OF PREM. Data signals still appear at the output of the PREM but they are not transferred since the write-clock (write data strobe) pulses are removed.

A counter is now started to count four spaces from the end word to the CRCC code word. The CRCC generator is disconnected from the data during this time and pulsed once more. One more write-clock pulse is now applied 4 spaces after the 16th end word while the output of the CRCC generator is connected to the data output.

Again the counter waits for four spaces until it signals ADD LRCC to the tape control and thereby to the tape deck. The tape deck adds the LRCC but waits for the POM to read the LRCC before it turns off the tape motion. More on the read and write function under POM and POM control.

The POM and POM control also test the read data for accurate codes and if correct issues the signal NO CODE ERROR. The control is waiting for this signal. If it arrives, the first 16 logs circulating in the PREM will be erased.

If the NO CODE ERROR signal does not arrive when expected, a CODE ERROR memory is set and the PREM starts the block transfer of the same block all over. The only two (2) differences are that a signal is issued to the microcomputer called CODE ERROR and that the block data will be erased after this second transfer.

Search

Any search request, if accepted by the tape control, will produce the signal SEARCH IN PROGRESS. This is the verification that the tape control has accepted the request from the microcomputer and that the tape deck is considered not to be at POLDE anymore. Such a search must be completed in the shortest possible time and the tape must be returned to POLDE to be ready for the next block transfer.

The tape deck having no ramp down of the write head current requires an initial write forward command for about 1/3 sec. The reason for this is when a change from a write command to a search command is given, the write current is turned off which produces a mark on the tape. This mark would appear in the IRG area and interfere with the format.

The microcomputer can now take over control of the tape deck and move it back beyond the probable position of the requested block. Any motion requires a set up of the direction (forward, or reverse) and the speed (slow, fast). Execution takes place as soon and as long as the EXECUTE COMMAND signal stays on.

A signal is fed back to the microcomputer that indicates when the tape deck is in motion. TD IN MOTION prevents signals from overlapping.

With the tape deck moving in slow forward the microcomputer will read the data that enters POM. The microcomputer will look for the log with a time and date word which is equal or close to the requested date for which the search was started.

The microcomputer will then send the signal REQUEST DATE DETECTED to the tape memory system as soon as it finds the right date. The POM will now capture this block of data.

Before using the data and printing it out on the teleprinter, the microcomputer has to take the tape deck back close to POLDE. This can be done at slow or fast speed. Finally, the microcomputer switches the tape deck to slow forward. The tape deck on its own will overshoot POLDE by 10 inches and switch to the RECALL LAST BLOCK mode which returns the TD to the right position just beyond POLDE.

Removing the last search control signals finally completes the TD part of the search.

The post tape memory will always indicate or mark the requested log to the microcomputer for first read out. This signal is called POM or PREM LOG FRAME.

Should command signals cause the tape deck to enter the BOT region the tape control will override the microcomputer signals and return the tape to the BOT mark. The status load point is generated by the tape deck when the BOT mark passes by or is at the photo-sensor position. The tape control recognizes this, removes the REVERSE command if still applied, and after proper time delays returns the tape to the BOT mark. No further action will be taken if the microcomputer insists on a REVERSE command at this point.

The signal END OF TAPE is given when the EOT range of the tape is entered. This means, of course, that the reel should be exchanged as fast as possible. No fast motion is executed in the EOT range to avoid accidental loss of tension. Should this ever happen, the following rule should be followed to gain manual tape deck control:

Thread tape back on take off reel. Wind 6-8 windings on take off reel. Push load button. Push rewind button. Continue with normal reel-change procedure.

Post Tape Memory Control (POM Control)

The POM control responds primarily to signals from the tape control and operates the POM. Several functions can be distinguished.

The SHIFT SIGNAL SWITCH applies one of 2 possible shift signals to the POM shift register.

The READ CLOCK SIGNAL (RDS) from the tape deck is only applied if a block of data is to be transferred from the tape to the POM.

Criteria for switching to the READ CLOCK SIGNAL are:

(1) The tape must be in slow forward.

(2) A search must be in progress, and

(3) No block was captured yet during this search.

The exact timing of the transfer from SR SHIFT to TAPE READ CLOCK is not critical since no data need be preserved yet.

Data is now transferred into the POM and looked at by the microcomputer as it appears at the POM outputs. No READ CLOCK pulses are generated by the tape deck while the inter record gap passes the read head. The resulting lack of shift pulses for 48 ms will cause all data from the previous block to fade away in the dynamic shift register of the POM.

The picture changes however if the signal REQUEST DATE DETECTED was received to capture the block that is being transferred.

The switch back to the SR SHIFT will occur about 200 us after the read clock signal that shifted the last data word but before the CRCC word.

The exact timing of this pulse is important to avoid errors in proper tracking of the block, to the end of the POM. The exact moment of transfer is the trailing edge of the SR SHIFT signal. This switch is not seen by the register since both shift signals are negated (positive) at this time.

The shift signal will not switch again until another search is called for and at least one signal NEXT POM LOG is applied. This signal releases the memory which indicates the capture of a block in the POM.

The CAPTURE BLOCK MEMORY terminates the block transfer and also prevents any more reactivation of the shift signal switch until reset.

Reset or enabling of the next possible block transfer is done with a signal NEXT POM LOG after a search is terminated, which includes returning the tape deck to POLDE.

A block signal generator which detects essentially a missing pulse in a train of pulses defines the presence of data (BLOCK PRESENT signal) and the presence of the CRCC word.

This signal is important in deciding when to change from the TAPE READ CLOCK signal to the SR SHIFT signal. The block signal generator determines also

when to sample the CRCC and LRCC circuits for a code error.

Advance Block to End of SR - Timer

Part of capturing a data block in the POM is to erase any other data that could be left in the shift register from previous times or signals.

A timer is used to accomplish this function. The BLOCK PRESENT signal is used to reset the timer during an IRG if the CAPTURE BLOCK MEMORY is still off. The first word of the next block starts the timer at the rate of words being read from the tape.

If the CAPTURE BLOCK MEMORY signal is turned on, the counter keeps counting beyond the end of the block although the count rate changes from the TAPE READ CLOCK to SR SHIFT. The POM changes from write to erase during this phase of the transfer process. The counter reaches finally a count equal to the total length of the shift register (2048) at which time the count is stopped and the shift register is connected for recirculation. The requested block was captured with all other data being erased.

Code Error

Data read from the tape is checked for three parity conditions.

(1) Vertical parity (IC 74180). Every word excluding the CRCC and LRCC words is checked for vertical odd parity. Any parity error will be remembered in a flip-flop until the end of the block when all code errors are sampled.

(2) Longitudinal parity (IC 8502). Each channel or trace on the tape is tested for an even longitudinal block parity. No error is indicated by an LRCC MATCH signal.

(3) Cyclic parity (IC 8500). The cyclic redundancy code was added as an additional word following the block data at 4 spaces. The same data is used again to form a new CRCC code with the difference of using also the taped CRCC word. A CRCC match is obtained if no error occurred.

A signal NO CODE ERROR is generated at the end of the block. This signal is always produced whenever the tape deck operates in slow forward. The PREM control uses this signal to determine the proper transfer of a block.

Log Number Detector and Counter

A signal must be generated to enable the microcomputer to pick the requested log out of the block of data circulating in the POM. This is done by using the lead and end word of each log. Each lead word that enters the POM sets a counter in the POM control. The counter is reset through the sequence 1 through 16. This presetting is interrupted when the signal CAPTURE REQUEST BLOCK arrives. The counter memorizes now the log that was recognized by the microcomputer as the one containing the requested data.

The lead word in question (in the counter) is now compared with all the data rotating in the POM. A signal is generated each time the same lead word passes by the POM output. A FF memory is set at that time that permits the SR SHIFT pulses to be transferred to the microcomputer. The microcomputer transfers now the requested log to its own memory. The end word of that log terminates the log transfer.

The log counter is advanced by the microcomputer by means of the signal NEXT POM LOG. This signal may be issued one at a time or in groups thereby skipping several logs. The count sequence is 1 through 16-1, 2, etc.

The POM control contains also the clock generator which generates signals of various frequencies for synchronizing, timing and counting purposes.

Post Tape Memory (POM)

The POM or Post Tape Memory provides storage for up to 2048 words of data which were retrieved from the tape deck during a search. This data is made available to the microcomputer on a per log basis. The POM board provides also a switch that permits the data output to transfer the last log in the PREM to the micro-computer. Finally, the POM has the major task of testing the read data particularly during a write cycle for correct error codes.

Data being read from the tape deck is fed to the POM. This includes a READ CLOCK signal. These signals are fed to various LSI circuits on the POM board to test for 3 individual error codes. All test results are then combined at the end of the block into a single NO-CODE-ERROR signal issued by the POM control.

The vertical parity test (IC 74180) compares the parity of each 8 bit word plus the parity bit with required odd condition. Any even parity result will be recognized as a VP error. The 2 code words at

the end of the block (LRCC, CRCC) also have odd parity since we have an even number of data words.

The longitudinal parity check (IC 8502) applies to the whole block of data. Each channel or tape track must have even parity when tested over the full data block length including the 2 code words. Any deviation will be recognized as an LRCC error. The LRCC code re-established, by the way, the reset (or erase) state of the magnetic flux on the tape.

The cyclic redundancy code (IC 8500) can be compared to a diagonal parity test. It also includes the whole block but not the LRCC word. Any CRCC error is also registered and used to obtain a parity error signal.

The parity error test is done every time the tape deck operates in slow forward. During block transfer, the parity error test is used to terminate the block transfer or to cause a repeat. During search or any time we retrieve data, the test has no active function but monitors the correctness of the data.

A Combination of 3 NAND

Input gates permit to write data into the dynamic shift register, erase data, or recirculate it. Data is written into the POM at a speed that is determined by the READ DATA STROBE signal from the tape deck (10 KHz rate). There is no clock signal during the passage of the inter record gap. The dynamic shift register however, requires shift pulses at a minimum rate of about 5 KHz. Data read into the POM will, therefore, fade away during the IRG.

All logs are checked by the microcomputer for containing the right time and date during the search. Once the right log is recognized, a signal from the micro-computer to the POM control will indicate that the block presently being read should be captured.

The POM will then switch the shift register from the READ DATA STROBE to the SR SHIFT signal at the end of the data before arrival of the CRCC word of that particular block.

The data block is then advanced to the end of the SR at this new speed while the gates are switched to feed "0"s behind the block into the SR. When the first word of the block arrives at the end of the SR, those gates are switched to the recirculate mode. The data block is now captured and circulates in the memory. Data is now ready to be transferred to the

microcomputer log by log. This function is controlled by the POM control. Switches at the output of the POM permit to either let the microcomputer look at the data in the POM, or at the last log in the PREM.

A vertical or word parity generator established again the parity bit for transmission to the microcomputer. One might remember that the parity bit is not stored in the POM (as well as the PREM).

7.2.4 CRT Subsystem

The CRT subsystem consists of either one or two TV monitors as specified on the order. The monitors are driven by circuitry on up to five printed circuit boards which, in turn receive periodically from the microcomputer, encoded data containing information that is to be displayed. Each monitor includes a 13-inch CRT tube with medium-persistance, yellow-green phosphor and the conventional circuitry necessary to accept a standard video signal over a coaxial line. For simplicity in the following discussion, the term CRT will be used to refer to the monitors including both the CRT tube itself and circuitry on the monitor chassis. When only one monitor is supplied, the two video signals are switched external to the driver boards via an RF relay controlled from the Trend Select panel.

A stream of up to 76 bytes of parallel data plus parity is sent every 1/3 second via a DMA channel from the microprocessor board to the group of three PC boards, located in the same rack with the microprocessor board, that interface with the CRT's. Since the data sent by the microcomputer is asynchronous with respect to operations taking place within the CRT board group, it is accompanied by suitable control signals to assure proper entry of new data into a buffer shift register located on the master board. The buffer shift register makes available its data every 1/60 second to the bar board for updating the information transmitted to the CRT and to the trend board for updating the information transmitted to the trend CRT.

The bar CRT is used to display two columns of horizontal bar graphs with up to 25 bar graphs in each column and with each column occupying somewhat less than 50% of the width of the screen. The bars to be displayed and their location on the screen are specified by the customer. Bar lengths correspond to instantaneous values (within 1/3 second) of the data they represent. This results from the fact that the microcomputer system samples all monitored data inputs every 1/3 second and sends on to the master board those data to be displayed on the bar CRT. New values

received in the buffer on the master board will appear on the bar screen within 1/60 of a second.

The trend CRT is used to display time-amplitude line graphs of up to 20 data quantities selected from those dispalyed on the bar CRT as specified by the customer. Each line corresponds to one hour's immediate past history of the data is represents. All lines consist of 80 points each and they are all updated every 45 seconds from current data then held in the buffer shift register on the master board. Push-pushbuttons on a trend select control panel located near the trend CRT permits the viewer to select one or more lines in any combination to appear on the screen at one time. Pushing a button merely permits or inhibits the display of that line on the screen and does not affect the memory that stores that line's history.

The trend CRT is used also to display two micro-computer-generated time - amplitude graphs called slope lines. There are on the trend select panel two additional sets of pushbuttons. One set can be used by the viewer to request the microcomputer 1) to generate and display a line of specified slope having a present value corresponding to one of the data lines and 2) to thereafter update that line along the specified slope as though it were another data point. The other set of pushbuttons can be used to request a similar line having a present value corresponding to another of the data lines.

Communcation among the basic elements is as follows:

(1) Every 1/3 second the microcomputer sends out 76 bytes of parallel data plus parity at a 417 KHz rate over the DMA bus (lines DD0 to DD7 plus DDP) to the master board along with an advance signal that leads the data by 150 microseconds and a MASTER-RECIRCULATE signal to allow the data to enter the buffer and a clock signal to clock the data into the buffer. A parity error occurring during transmission will cause the associated data point not to be entered into the buffer and will cause a data parity error signal to be sent back to the microcomputer at the end of the transmission.

(2) Every 1/60 second, during the vertical blanking period of the display cycle, the buffer shift register copies its data at a 1.25 MHz rate onto the buffer shift register output bus over which the data is sent to both the bar and trend boards.

At the same time, clock and recirculate signals are sent to the bar board to permit entering the new data into the bar display shift register.

(3) During each vertical unblanked period of the bar display cycle, the data in the bar display shift register is shifted out as required, translated into electrical signals representing bar length and combined with a composite synchronizing signal to produce a video signal which is sent via a coaxial cable to the bar CRT which may be located up to 200 feet away. Electrical signals containing graticule information generated on the master board are also included in the video signal sent to the bar CRT.

(4) Every 45 seconds, during the vertical blanking period of the current display cycle, while the buffer master shift register is copying its data onto the buffer shift register output bus, update circuitry on the trend board is enabled by a counter on the master board and it selectively updates the line storage shift registers located on the trend board. (One shift register, eight bits wide and 80 bits long is required to store each trend line.) That is, every 45 seconds, as the data stream appears on the buffer shift register output bus, each trend line shift register is updated with the particular byte in that data stream for the particular data line that it has been assigned to store.

(5) During each vertical unblanking period of the trend display cycle, the data in the trend-line-storage shift registers is shifted and organized by the control circuitry to produce electrical signals representing the lines and combined with a composite synchronizing signal to produce a video signal which is sent via a coaxial cable to the trend CRT which may be located up to 200 feet away. Electrical signals containing trend graticule information generated on the master board and on the bar board are also included in the video signal sent to the trend CRT.

Standard 525 line interlaced scan is used. A sync generator located on the master board provides the required composite sync signal for both displayes as well as horizontal and vertical drive signals which are used to synchronize the data with the composite signal.

In addition to data and graticules, present time and date information is displayed on one of the displays. This information is sent from the microcomputer to the

master board every 1/3 second as a series of bytes after the data bytes in the same DMA transmission used to send data. This information is converted on the time and date or alpha numeric board, whichever is used, into an electrical signal which is added to the video signal that goes to the monitor.

The CRT subsystem operates from a 10 MHz clock located on the master board where subclocks of 2.5 MHz, 1.25 MHz and 1.67 MHz are derived.

When electronic generation of the legend and scale markings are provided, the signals are generated on two boards namely alpha- numeric and message.

The alpha-numeric board contains two character generators, EPROMS, and necessary circuitry for adding the legend and scale markings to both the bar and trend displays. In addition, time and date will be displayed on either the bar or trend display depending on the system makeup.

To accomplish this, each display is divided into small segments and each segment is assigned an address, whereby the character for that position may be encoded and stored in a PROM. This is done by the use of 3 2708 1K byte EPROMS per display. The vertical position of the character within each segment is also programmable and may take one of four possible positions. This portion of the program is stored in a 256 x 8 EPROM (1702A).

A counter synchronized by the CRT's vertical and horizontal drive signals provide the addresses for the character and position PROMS. When addressed, the PROM data is moved into latches and applied to a 7 x 9 raster scan static character generator with ASCII code conversion. The data is then decoded and the character outputted to an 8 bit shift-register where the character is serially fed to the appropriate video amplifier on the master board.

In addition to the legends and scales, time and date is added to one of the displays. The time and date information is taken from the shift-register update bus every 1/60 second and stored in read/write memory. Proper control signals then routes the data to the proper character generator.

The message board contains a voltage regulator which supplies -5V required for the 2708 PROMS on the alpha-numeric board. This board also contains circuitry for generating some required synchronizing and routing signals used in conjunction with the alpha-numeric board.

The main purpose of the message board, however, is for decoding the 75th data word in the buffer shift-register to determine if a message is to be displayed. If so, it provides the necessary blanking and routing signals to cause one of the standard displays to be blanked and from one to eight messages stored in PROM, on the alpha-numeric board to be displayed in its place. The display of these messages is controlled by the software program. Not all systems utilize this feature. If used, the MI will list the messages and conditions under which each will be displayed.

Graticules

There are three hardware-implemented horizontal graticule generators located on the master board that can be used to divide the horizontal distance representing 100% bar length into 3 majors, 5 minors; 4 majors, 5 minors; and 5 majors, 5 minors. Any of the three graticules can be assigned to any of the 4 quadrants on the bar CRT by means of jumpers located on the bar board. Only the 5 major, 5 minor horizontal graticule is used on the trend. Its vertical distribution is controlled by counters located on the bar board.

Video Amplifiers

Video amplifiers for the two displays are located on the master board. The bar circuit video amplifier receives separate data and graticule signals from the bar board as well as legend signals from thealpha-numeric and message boards, if used. The trend circuit video amplifier receives a data signal from the trend board, a graticule signal from the bar board, and a time and date signal from the time and date board. If used, the alpha-numeric and message boards replace the time and date board. In each case, the signals are combined along with a composite sync signal from the sync generator to produce a composite video signal. In each case, potentiometers are available on the front edge of the master board to permit independent adjustment of the amounts of data, graticule, and legend signals that are used in making up the composite signal. The relative intensity of each can thus be adjusted.

CRT Screen Formatting

Fig. 6 shows how the bar CRT display area is positioned within the available screen area. Lines along the left edge marked left side blanking and top bottom divider, represent control signals generated from the scan line counter and are used to gate graticules and data areas on the left side of the screen. There are

corresponding lines along the right hand edge that are used for gating graticules and data areas on the right side of the screen. Transition points A, B, C and D on the left side and points A', B', C' and D' are selectable by means of a printed wire matrix on the bar board which is custom-connected via solder pins. The scan line counter is located on the master board. The horizontal counter measures horizontal position of the beam in microseconds starting at a point delayed beyond the end of the horizontal retrace period by an amount indicated by HD DELAY.

The scan line counter starts counting lines at preset line 232 at a point delayed beyond the vertical retrace period by an amount indicated by V_D DELAY.

Fig. 7 shows how the trend CRT display area is positioned within the available screen area. In this case there is but one large display area requiring similar, but simpler, control signals.

8.0 SOFTWARE

Software controls many tasks in the ATSI. Most of them are complex enough to require special programs and procedures. Each will be discussed briefly for better understanding of the system although a copy of the program will not be supplied and any modifications must be made by General Electric Company.

During troubleshooting, if the software is suspected, a spare set of PROM boards containing the program should be installed. If defective, return the boards to the factory for repair.

Interrupt procedures are used when the control software needs to be synchronized with external events or needs to be run at a known rate and are discussed first.

Interrupt 0

INT 0 performs the initialization of the ATSI on reset or power-up. When power is applied, this program will run for 1.5 seconds, during which time 3 initialization tasks are accomplished. First of all, the RAM is cleared, set equal to zero. Next, the CRT's bar and trend, must be cleared of random information. This is done with the assistance of procedure INT 1 to send clear data to the CRT subsystem until it's memory is cleared. The last task is to set-up key variables used to initialize special tasks the first time INT 2 is run.

Interrupt 1

INT 1 is used to output the slope lines to the trend CRT.

Once this interrupt is enabled, it executes at a rate of 60 times per second. On the completion, it disables itself and will not execute until enabled by the control program.

Interrupt 2

INT 2 is referred to as the main control program. Once initialization of ATSI by INT 0 is complete and the operator presses the start pushbutton, INT2 will run at a rate of 3 times per second. Now every 1/3 sec. the main control program will execute most of the control tasks. All the system analog inputs will be read. This is accomplished by the procedure MUX. Then the data from the real time clock is read using the procedure READCLOCK. Afterwards, the data required to be displayed on the CRT is transferred from the MUX input table, which is also the magnetic tape output buffer, to the CRT output buffer. Every other time the main control program is executed, the limits on selective analog signals are checked against set points by procedure LIMITCHECK.

Commands from the operator through the trend select panel for new slope lines are handled by INT 2. Once this input data is decoded, this procedure will calculate the new slope and with the assistance of INT 1 output the slope line. After the slope line has been outputted, INT 2 calculates and outputs every 18 seconds a new point. This is done to keep the line updated as time shifts it across the CRT.

Logging of data on the magnetic tape is also a task controlled by this program. It will look at time from the real time clock, alarm flag and status flag to determine the logging rate. The sensing of action requests from the operator through the teleprinter is part of the interrupt 2 software. Once the operator inputs the command, action to execute the request is started by this control program but may also require assistance from interrupt procedure 4 to complete reading and decoding the command.

The last main task executed by INT 2 is Auto-Log. Once every day at midnight, the control program will exercise the magnetic tape system by requesting a search for the first log in the last block recorded on the tape. Once the system has inputted the log, it will initiate the output of the log to the teleprinter.

Interrupt 4

INT 4 services data input from the teleprinter by the operator. Commands like RD, BOT and LDE have an input string of numbers to be read and decoded. The main control program, after acknowledging the operator command, will enable this interrupt routine to do the inputting and decoding of the data string. Once INT 4 has completed the decoding, it will, in the case of command RD, check the validity of the data. Also, if the

data for command RD was valid, it will calculate the rewind time for the tape.

Interrupt 5

INT 5 is the procedure that services the data being transmitted to the teleprinter. After the main control program has requested that data, either a message or log, be outputted this interrupt routine is enabled. Once enabled, data will be outputted character by character with the conversion from internal format to a useable code for the teleprinter, being done by this program.

Interrupt 6

INT 6, commonly known as the magnetic tape interrupt, is used during the searching for logs on the tape. Once enabled by the main control program, it will read the data from the tape at a rate of 10,000 characters per sec. It synchronizes with this data and examines the time and date on the logs for a match with that requested by the operator. Once the log has been located it is inputted to the processor and control is returned to the main control program. Also during the search, interrupt 6 weaves the main control program in as time dictates.

Procedure MUX

MUX is called when the main control program needs to input the analog points. It's task is to input the system signals and check for over-ranging. Once this has been completed, updating of peak speed is performed. The last but main task of MUX is the linearization on the thermocouple data. At the completion of linearization, control is returned to the calling procedure.

Procedure MSEC

MSEC is a program used when time delays are needed. When this program is called, it delays for 150 usec before returning to the calling program.

Procedure READCLOCK

READCLOCK is used to read the clock data (years, days, etc.) off the real time clock. Since the validity of this data is essential, data filtering and verification is done before returning to the calling procedure.

Procedure BARDATA

BARDATA is used to control the CRT DMA. When called, it will output the signal which lets the CRT subsystem place the microcomputer in a DMA HOLD state. Once the transfer has been terminated, this program returns to the calling procedure.

Procedure OUTBARGR

Outputting of data to the CRT subsystem is initiated by this program. To accomplish this task, it uses two other procedures BARDATA and MSEC. First an advance signal is sent, then BARDATA initiates the transfer to the CRT subsystem. Once the transfer is complete, OUTBARGR clears the DMA signals and returns to the calling program.

Procedure NEGNO1

NEGNO1 checks the value of slope line 1. When its value is positive, the program stores the slope value in the output table. When the value is negative, NEGNO1 will store a value to keep the slope off screen.

Procedure NEGNO2

This procedure is identical to NEGNO1 but operates on slope line 2.

Procedure TRUP

The task accomplished by TRUP is to update the slope lines, 1 and 2, when called by the main control program. TRUP is controlled by two flags which signify the slope lines now being displayed and their signs. Once the new slope points are calculated, all the CRT table is outputted by calling OUTBARGR. TRUP then returns to the calling program.

Procedure PAIRLIMIT

PAIRLIMIT checks paired analog inputs to their limits. The maximum differential between these pairs are checked and if in excess, an alarm is noted. Also on alarms, the alarming pairs value to be outputted to the CRT's are set to zero.

Procedure LIMITCHECK

LIMITCHECK is called by the main control program every other time it executes to check limits on all required analog points. With the assistance of lookup tables, the program checks alarms on vibration, shell metal temperatures, water detection temperatures, differential expansion, journal bearing temperatures, and thrust bearing plate temperatures. When checking paired values, this program calls on procedure PAIRLIMIT.

Procedure PCU

PCU is used by INT5 to control the formating of the teleprinter output. During printing, this program takes care of formating of the line feeds at table headings. Also when the output is completed, PCU will disable the transmitting printer's interrupt.

Procedure TTYCHECK

TTYCHECK is used by the main control program before using the teleprinter for output. TTYCHECK checks the availability of the teleprinter by interrogating hardware and if the teleprinter is not in used, reserves it.

Procedure TTYRELEASE

Once a program has completed its uses of the teleprinter, TTYRELEASE is called to release the teleprinter for uses by another ATSI.

9.0 MAINTENANCE

A recommended maintenance schedule is given in summary form in Table III. For further information on servicing the major components of the system, refer to their separate publications which are listed in the system TABLE OF CONTENTS.

TABLE III

RECOMMENDED MAINTENANCE SCHEDULE

RECOMMENDED SCHEDULE	PROCEDURE
Every 3 months	Shaft Vibration Detectors - Inspect for wear of rider tip. Oil upper probe bearing with several drops of light machine oil. (No maintenance required for bracket mounted detectors.)
Every 3 months	Cabinet Air Filters (44A235705-016)- Check and clean as required with hot water and detergent. Wring dry as possible.
Each Tape Change and Each Time Recorder is "Off Line" for Any Reason	Tape Recorder - Refer to separate publication.
Every 6 months	Teleprinter - Refer to separate publication.

The control cabinet should be opened only when necessary for service or inspection and then only by qualified personnel. The surrounding area should be kept clean and dust-free at all times, especially during intervals when the doors must be open for servicing.

Maintenance Records

Most users develop a procedure for keeping records showing maintenance routines including dates of lubrication, details of down time, results of various checks, and information concerning service calls. Such records are strongly recommended.

10.0 TROUBLESHOOTING

10.1 ANALOG SECTION

Troubleshooting instructions for the individual measurement channels will be found in the instruction publications for the various measurement channels.

The following instructions relate only to the analog section, particularly the power supplies, and will cover some of the most common troubles which may be encountered, their causes and remedies:

I. TROUBLE: Power pilot light does not come on when POWER switch on the power supply is thrown ON.

A. PROBABLE CAUSE: No AC power is reaching the power supply.

1. REMEDY: Check and correct AC voltage-supply lines to the power supply.

B. PROBABLE CAUSE: Blown or open fuse in the power supply.

1. REMEDY: Remove fuses from the front of the power supply. Test and replace if open.

II. TROUBLE: In step 5 or 6 of the PRELIMINARY TESTS AND ADJUSTMENTS, the output voltage of one or both of the DC power supplies is zero.

A. PROBABLE CAUSE AND REMEDY: See TROUBLESHOOTING section of the publication for the DC power supply.

III. TROUBLE: In step 5 or 6 of the PRELIMINARY TESTS AND ADJUSTMENTS, the output voltage of one or both of the DC power supplies is not within limits of 15.0 $\pm$ 0.3 volts.

A. PROBABLE CAUSE: Voltage adjustment on dc power supply changed or disturbed from its original factory setting.

1. REMEDY: Readjust to 15.0 volts. (Do not disturb the adjustment if voltage is within the ± 0.3 volt limits.)

IV. TROUBLE: Time delay relay will not pick up when POWER switch on the power supply is turned ON.

A. PROBABLE CAUSE AND REMEDY: TRIP switch on the power supply is in the DISABLED position.

1. REMEDY: Turn TRIP switch to ON.

B. PROBABLE CAUSE: Vibration amplifier boards, vibration filler boards, vibration alarm boards, alarm filler boards, or trip board not in place.

1. REMEDY: Refer to the internal connection diagram furnished with the system instructions, and see that all circuit boards are in their places. Check that all boards are pushed all the way in as far as they will go, to be sure they are fully mated in their receptacles.

10.2 DIGITAL SECTION

The troubleshooting techniques described below are intended in most cases to trace the source of the problem to the particular board involved so that it can be replaced with a spare board. Defective boards should be returned to the factory for repair. Before replacing any board, first check that all boards and connectors are seated properly.

CAUTION

TO AVOID DAMAGE TO THE PRINTED WIRING BOARDS, IT IS NECESSARY THAT POWER BE TURNED OFF THE DIGITAL SECTION BEFORE BOARDS ARE REMOVED FROM, OR INSERTED INTO, THE LOGIC MODULES.

Data Display Problems

If data appears to be incorrectly displayed on both CRT's, the cause may lie in the transmission of data from the microcomputer to the master board. Incorrect transmission due to the absence of advance and re-circulate signals should cause alarm relay K1B to pick up. To determine whether the microcomputer is operating, turn on the teleprinter and request present entry (PE_) information. If this operation is normal, replace the master board. If operation is not normal, make further checks at the control cabinet as follows: Observe the indicator lights on the faceplate of the diagnostic board; they should be flashing at a rate of 3 Hz. If the lights are not flashing, reset the microcomputer by the reset switch at the microprocessor board. If the lights still do not flash, replace the clock board; the 1/3 second interrupt signal (3 HZI) is generated on this board. If the lights still do not flash, replace the I/O board; the interrupt circuitry is located on this board. If the lights still do not flash, replace the microprocessor board.

If data appears correctly on the bar CRT but is absent or incorrect on the trend CRT, first check to see if trend data can be controlled by the trend select push-buttons. If not, replace the trend board. If still incorrect, replace the trend select board.

If no legend or scale markings appear on the display but both the bar and trend data is correct, replace the alpha-numeric board. If the problem is not corrected, replace the message board then the master board.

If the microcomputer is operating and the PE_ data is zero for all points, replace the A/D board. If the data is still zero, replace the microprocessor board. If only certain groups of data registers zero, replace the MUX board associated with those signals. If the data displayed appears normal but no time and date is displayed, replace the time and date board or the alpha-numeric and message boards, whichever is used.

Graticule Display Problems

If data is correct on the bar CRT but graticules are missing, replace the bar board and/or the master board.

If graticules are missing or incorrect on the trend CRT, replace the bar board since gating for the trend graticules is on that board.

Data Recording or Retrieving Problems

If the PE_ and display data appears normal but data is not being recorded on tape, first check that the TAPE READY light on the tape control board is ON. If it is ON, check to see if the system responds to a search request. If it does not, replace the tape control, PREM, and POM boards, one at a time until the problem is corrected. If the problem is still not corrected, replace the PROM board(s). If the TAPE READY light is not ON, the problem is probably on the tape control board or in the tape deck.

Printout Problems

If the PE_ and RD_ printouts have missing or scrambled data, but the same parameters displayed on the CRT are normal, replace the PROM board(s). If the problem still exists, replace the I/O and microprocessor boards in sequence until the problem is corrected. If still not corrected, check for loose connections between the cabinet and the teleprinter. Finally, check the teleprinter itself.

If the PE_ printout is normal but the RD_ printout is not, replace the POM and POM control boards in sequence. If the problem still exists, check the tape deck.

11.0 REPAIR AND REPLACEMENT PARTS

The following spare parts are shipped with the ATSI system:

5 - spare fuses for power supply (Analog Section), 5 ampere, 125 volt, slow-blowing, Littel-fuse' 313005 or Bussmann MDX5.)

5 - spare fuses for test panel, 1/100 ampere, 250V, 3AG size, Littel-fuse 312.010 or Bussmann AGC 1/100.

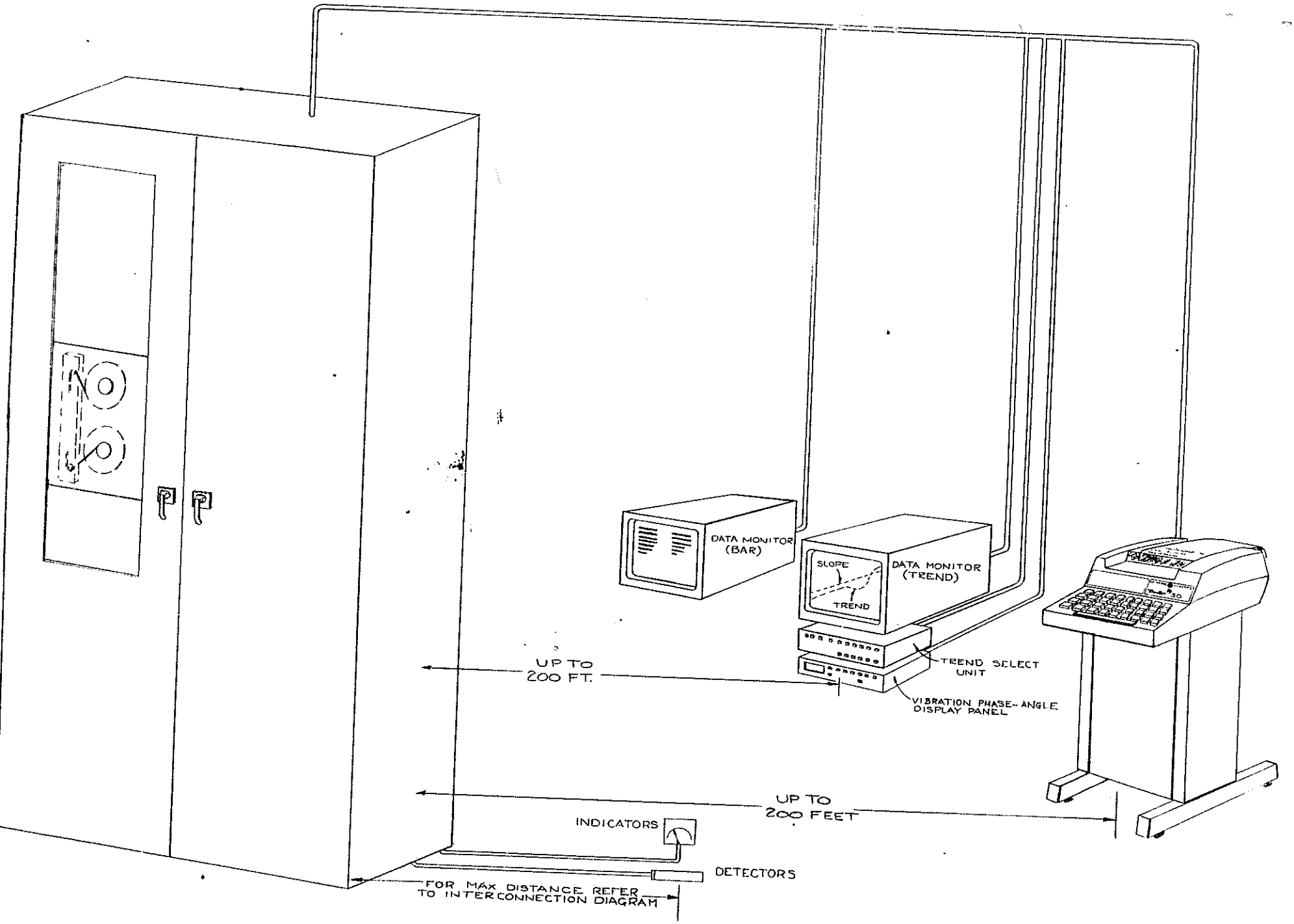
5 - spare fuses for test panel, 1/32 ampere, 125V, "Microfuse" Littel fuse 273.031.

4 - test leads for test panel, 4 feet long, two red and two black.

Any additional spare parts required may be ordered through the nearest General Electric Installation & Service Engineering office. Specify the drawing number or description of the parts, and also specify the system catalog number, which is given on the nameplate inside the rear door of the cabinet.

In the case of trouble with one of the printed wiring boards, it is recommended that a replacement board be installed. In general, it is not recommended that any attempt be made to repair printed wiring boards in the field. It is good practice to have available one spare printed wiring board of each type used in the system.

Printed wiring boards may, if necessary, be returned to the factory for repairs. For information on repair service, contact the nearest Installation and Service Engineering Office of General Electric Company.



TYPICAL ATSI SYSTEM

FIG. 1

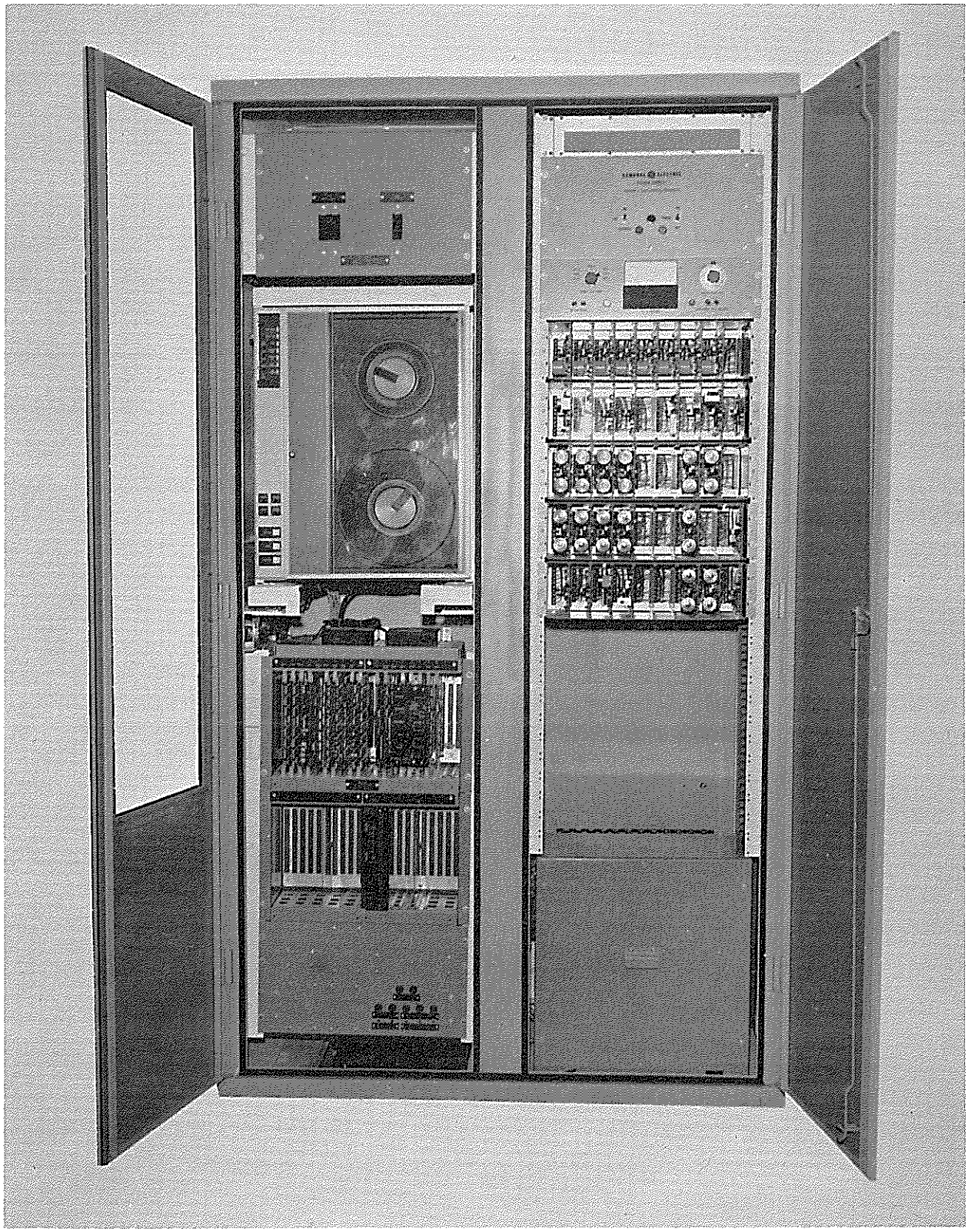


Figure 2 - Typical ATSI Cabinet - Front View

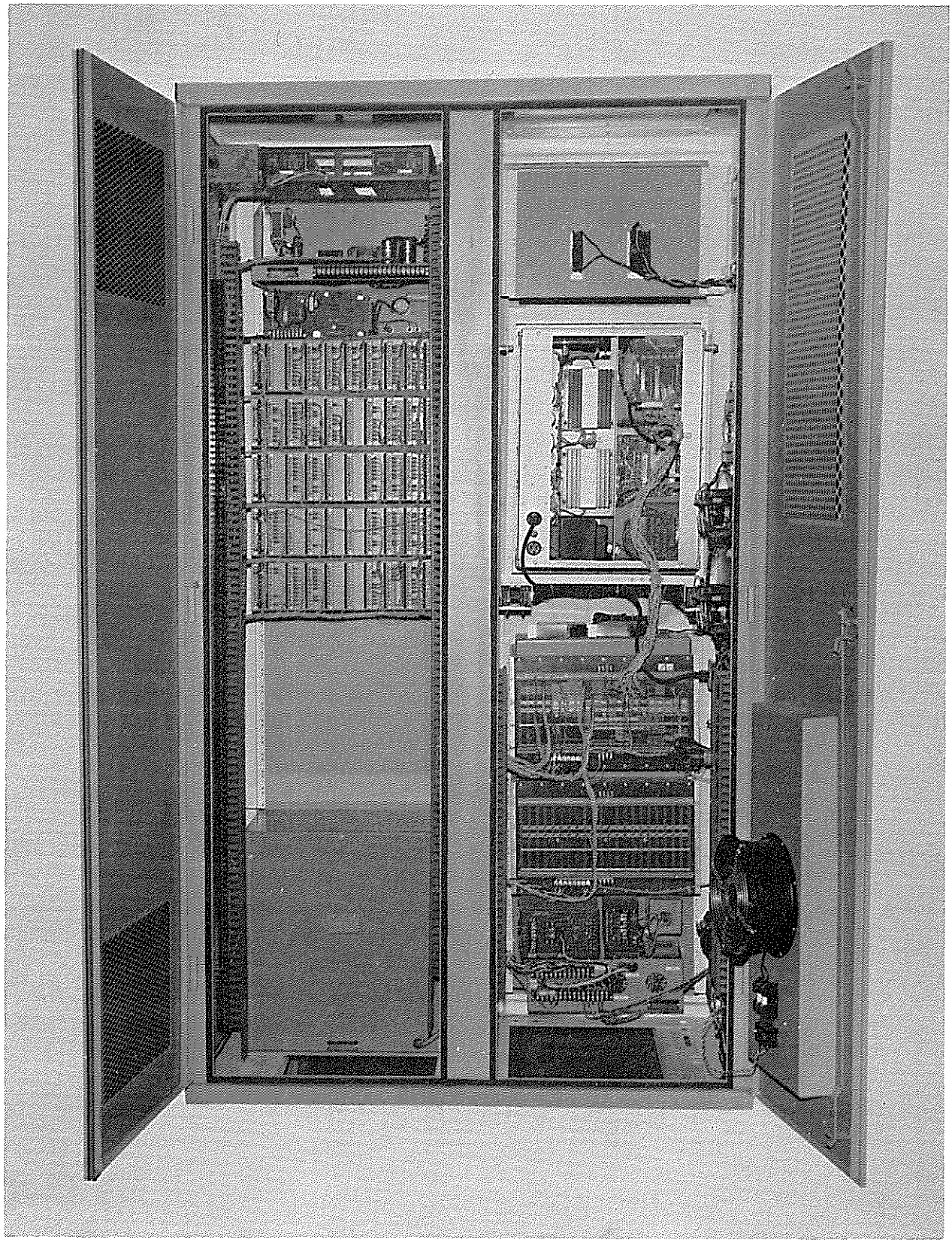


Figure 3. Typical ATSI Cabinet - Rear View

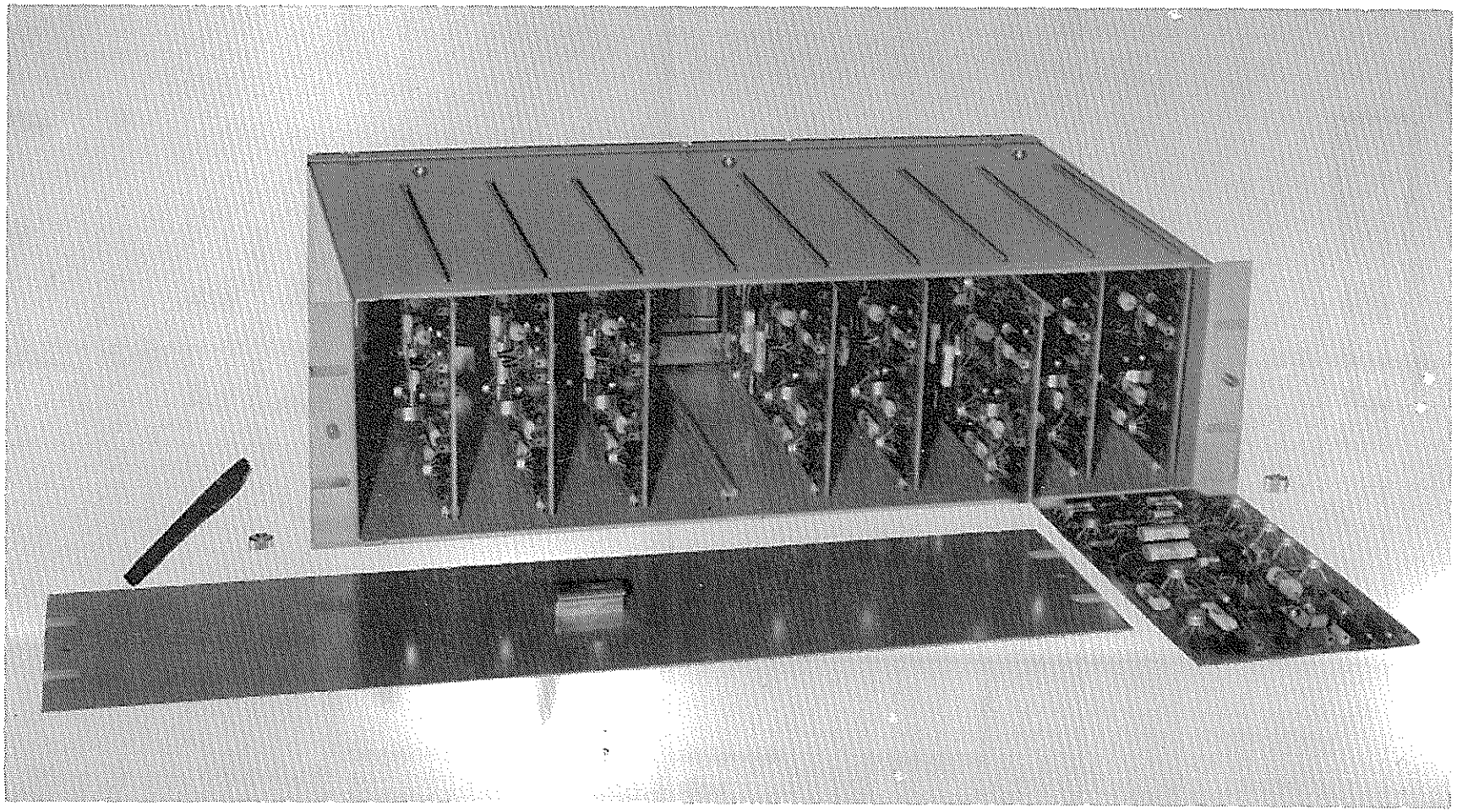


Figure 4. Typical Analog Module With Circuit Boards
(Cover Removed)

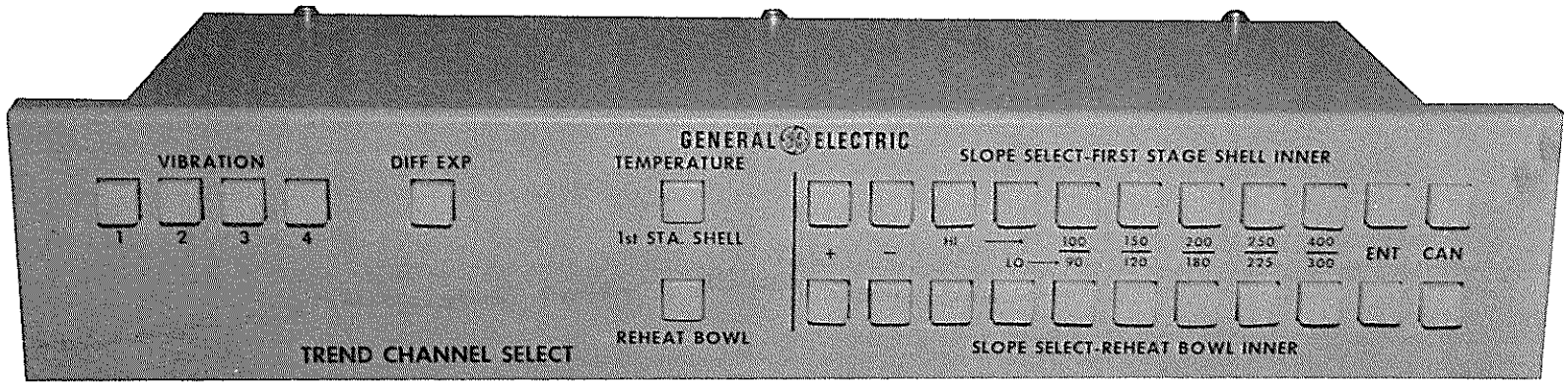
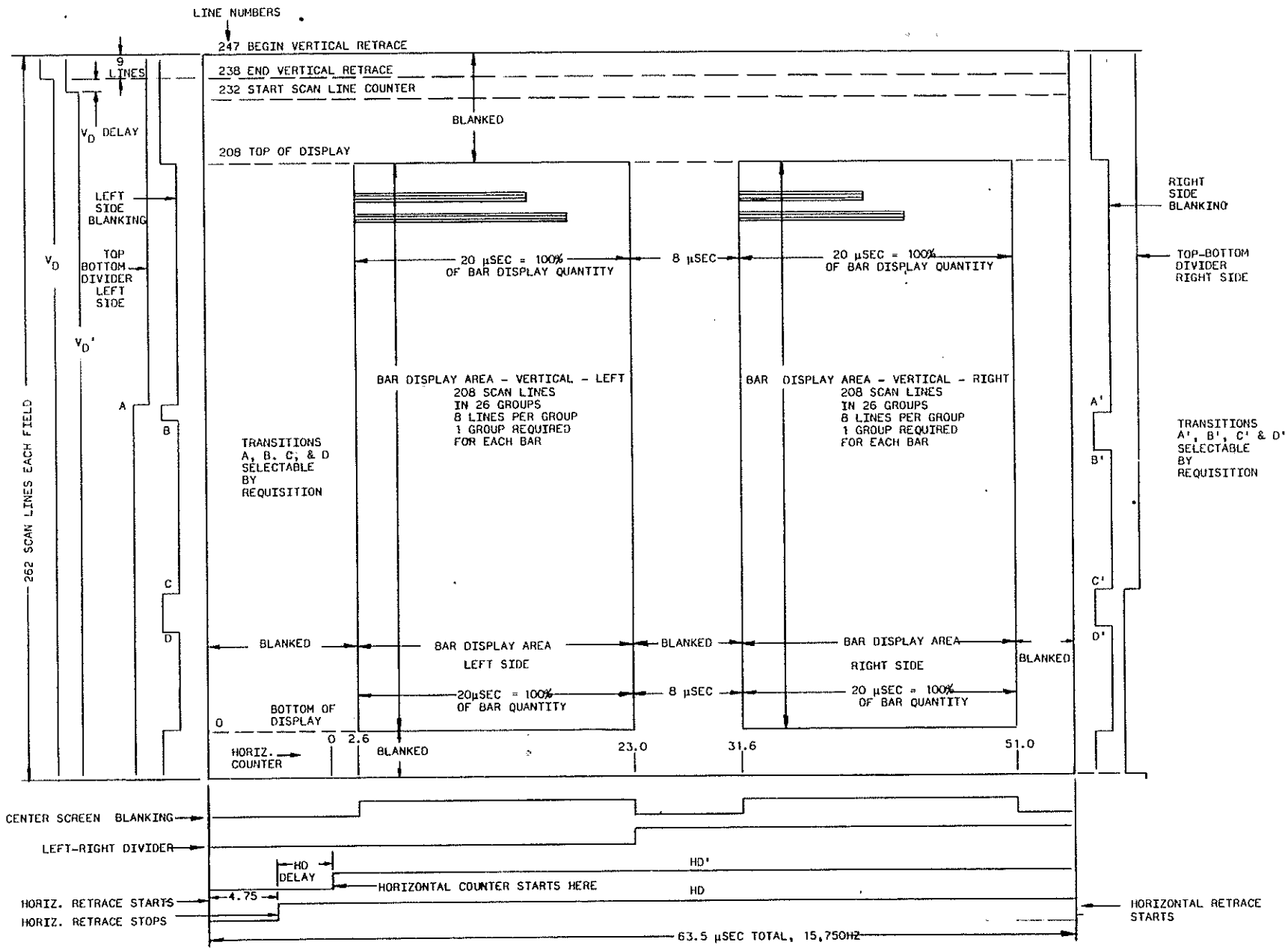
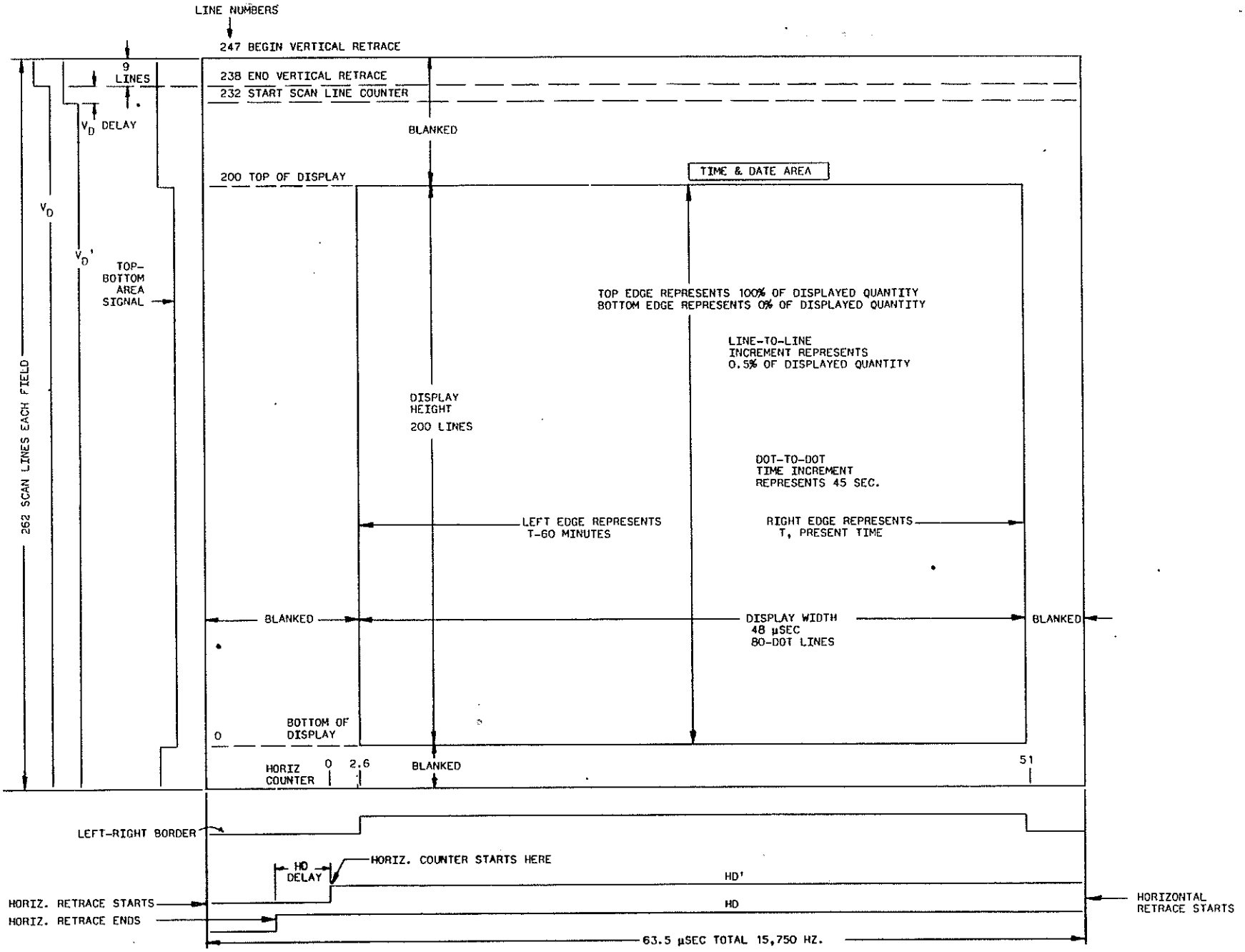


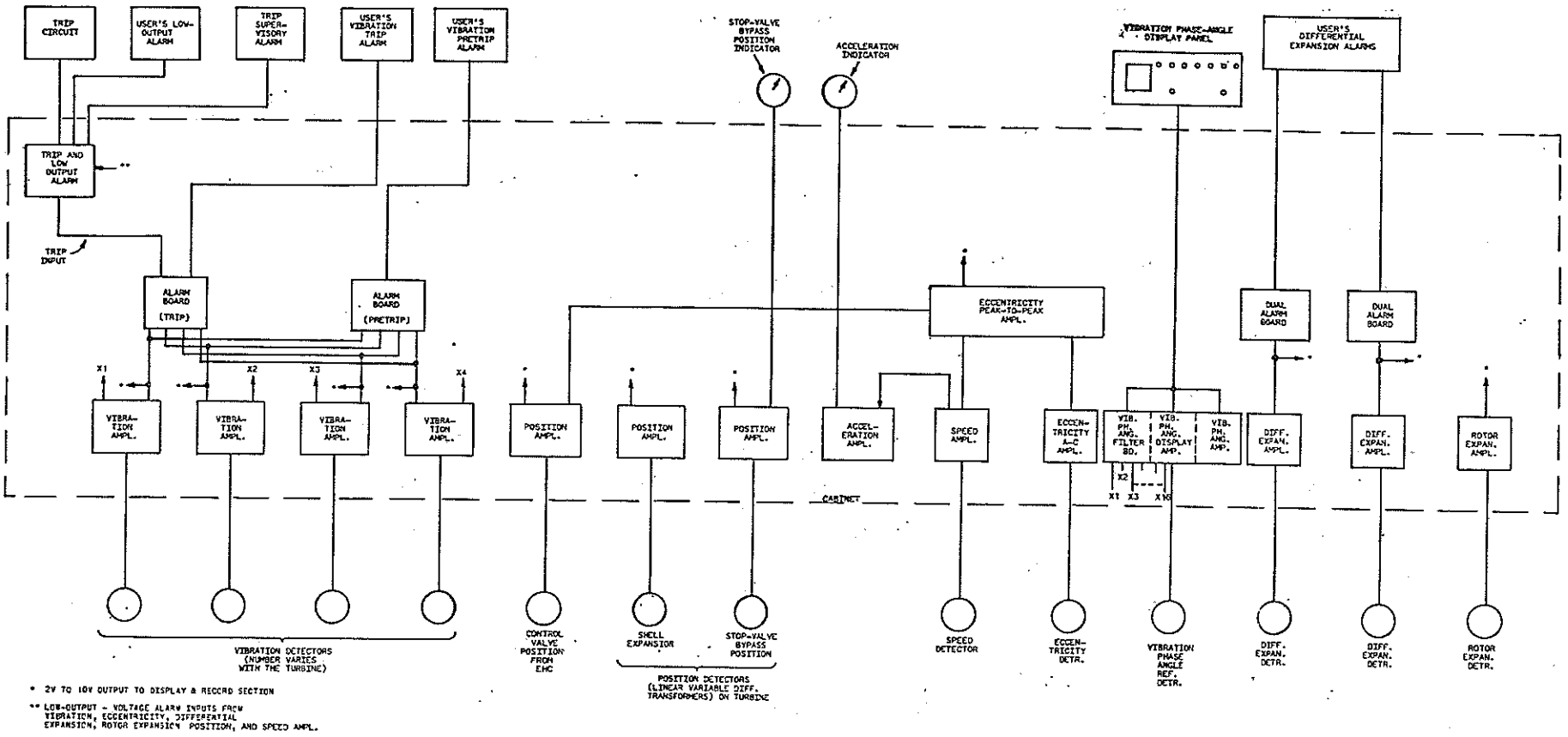
Figure 5. Typical Trend Select Unit



TYPICAL BAR CRT DISPLAY FORMAT
FIG. 6



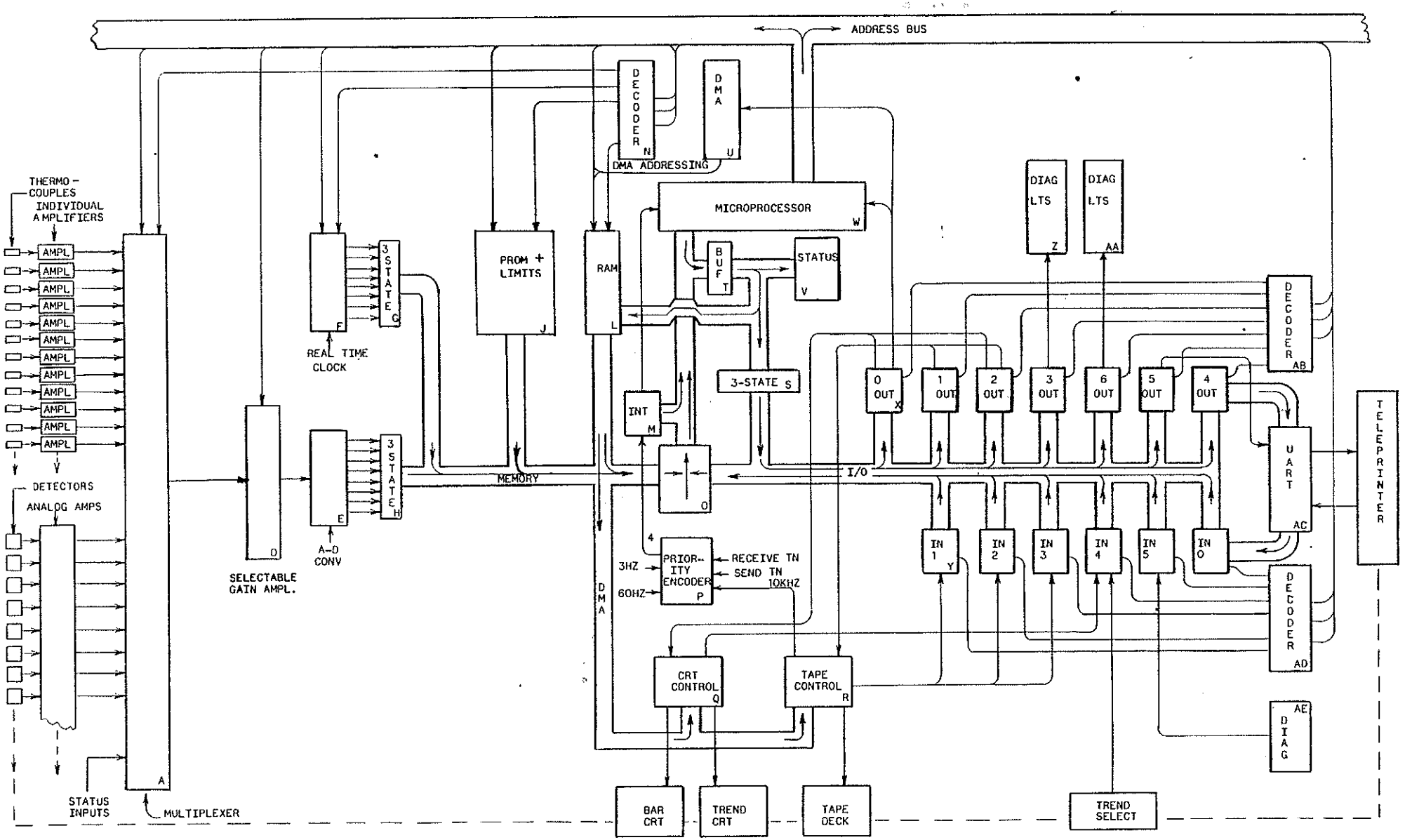
TYPICAL TREND CRT DISPLAY FORMAT
FIG. 7



* 2V TO 10V OUTPUT TO DISPLAY & RECORD SECTION

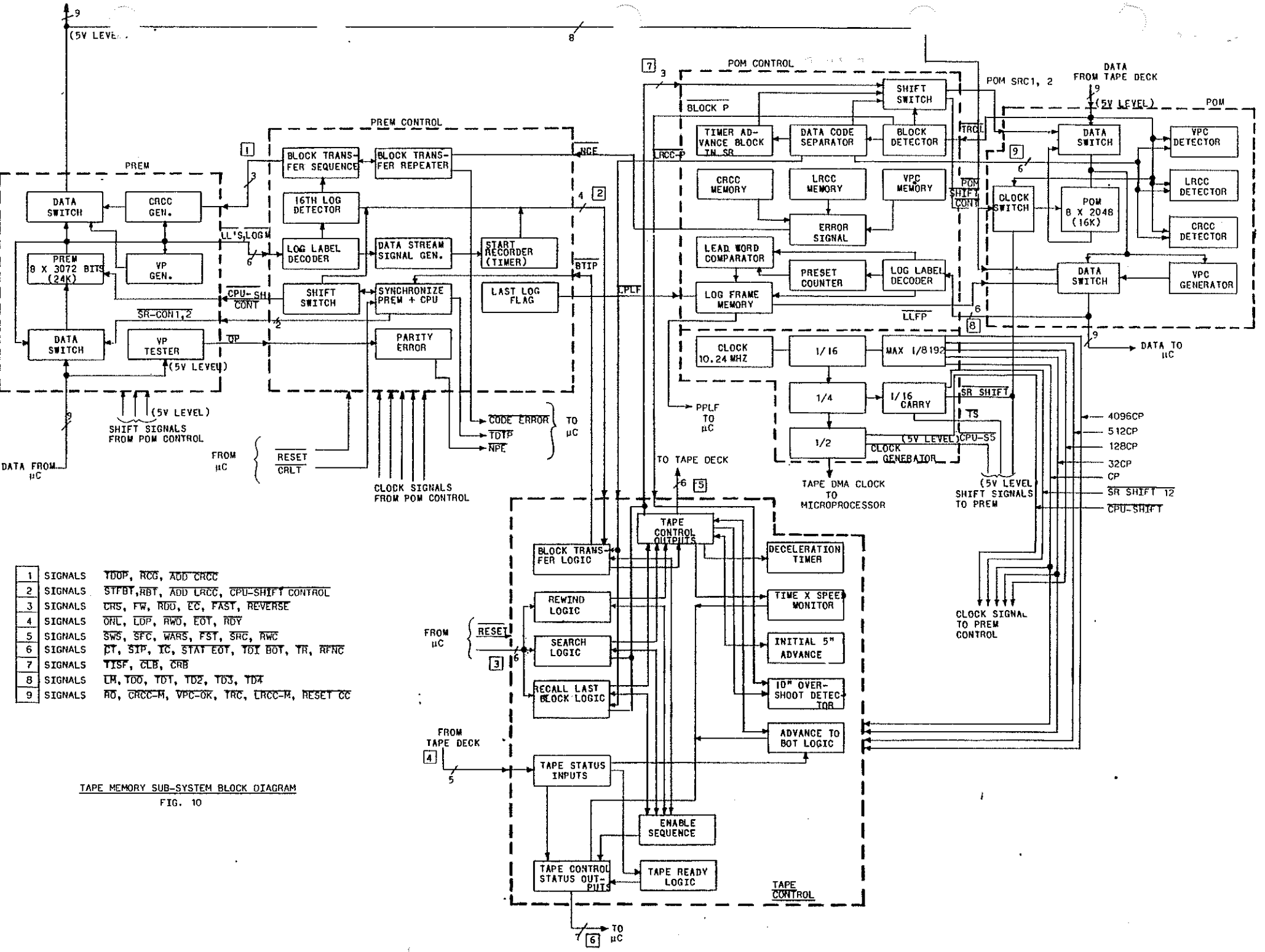
** LOW-OUTPUT - VOLTAGE ALARM INPUTS FROM VIBRATION, ECCENTRICITY, DIFFERENTIAL EXPANSION, ROTOR EXPANSION POSITION, AND SPEED AMPL.

FIG. 8 BLOCK DIAGRAM OF A TYPICAL ANALOG SUB SYSTEM



TYPICAL SYSTEM BLOCK DIAGRAM

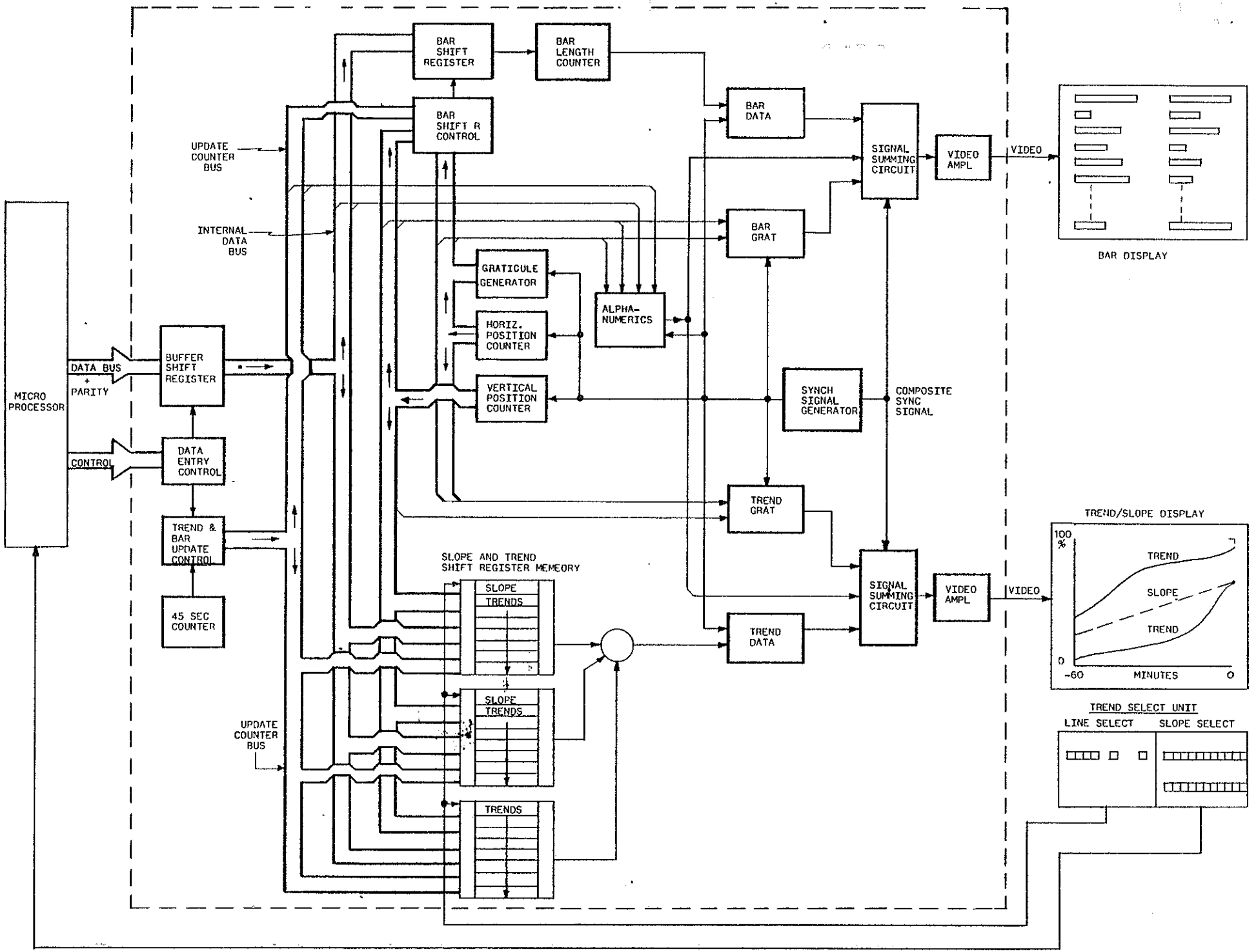
FIG. 9



1	SIGNALS	TDOP, RCG, ADD CRCC
2	SIGNALS	STFBT, RBT, ADD LRCC, CPU-SHIFT CONTROL
3	SIGNALS	CRS, FW, RDD, EC, FAST, REVERSE
4	SIGNALS	ONL, LOP, RWD, EOT, RDY
5	SIGNALS	SWS, SFC, WARS, FST, SRC, RWC
6	SIGNALS	CT, SIP, IC, STAT EOT, TDI BOT, TR, RFNC
7	SIGNALS	TISF, CLB, CRB
8	SIGNALS	LM, TDO, TD1, TD2, TD3, TD4
9	SIGNALS	RD, CRCC-M, VPC-OK, TRC, LRCC-M, RESET CC

TAPE MEMORY SUB-SYSTEM BLOCK DIAGRAM

FIG. 10



TYPICAL CRT SUB-SYSTEM BLOCK DIAGRAM
FIG. 11